

BANGALORE ELECTRICITY SUPPLY COMPANY LIMITED
(Wholly owned Government of Karnataka Undertaking)

Email ID – gmdas@bescom.co.in

No: BESCOM/BC-19/GM(DAS)/2023-24/1083-1087

Office of,

The General Manager (DAS)
BICC-1, 23rd Main, 17th cross
2nd sector, HSR Layout
Bengaluru – 560102
Date: 11/10/2023

DETAILED WORK AWARD NO. BESCOM/GM(DAS)/DGM-5/2023-24/161

DATED 11.10.2023

To,
M/s. NCC LIMITED
NCC House, Madhapur,
Hyderabad - 500081

BY REGD. POST ACK. DUE

Sir,

Sub: Upgradation of DAS in Bengaluru City - reg

Ref:

1. Tender Ref No. BESCOM/BC-19/GM(DAS)/2022-23/07 dated 24.05.2022
2. Tender published in e-procurement portal on 30.05.2022
3. Proceedings of the Pre-bid meeting held on 09.06.2022 vide BESCOM/BC-19/GM(DAS)/AGM-1(DAS)/2022-23/CYS-27A dated 09.06.2022
4. Clarification to Pre-bid queries issued vide BESCOM/BC-19/GM(DAS)/DGM-3/AGM-1/2022-23/647 dated 20.06.2022 and uploaded in e-procurement portal on 20.06.2022
 - Annx-1 Addendum issued vide BESCOM/BC-19/GM(DAS)/AGM-1(DAS)/2022-23/645 dated 20.06.2022
 - Annx-2 Amendment issued vide BESCOM/ GM(DAS)/BC-19/2022-23/646 dated 20.06.2022
 - Annx-3 Clarifications to Pre-bid queries
5. Amendment to the tender enquiry issued vide BESCOM/GM(DAS)/BC-19/2022-23/738 dated 28.06.2022
6. Your offer to the tender through e-Procurement platform.
7. Technical bid opened on 15.07.2022.
8. Tender Scrutiny Committee meeting held on 19.08.2022.

W.M. Shashidhar
W.M. SHASHIDHARA SHIVACHARI
General Manager (DAS & TIC)

9. TSC proceedings BESCOM/BC-19/GM(DAS)/2022-23/CYS-71 dated 24.08.2022
10. Clarification on document submitted against bid enquiry vide BESCOM/BC-19/GM(DAS)/DGM-5/AGM-1/22-23/1164 dated 25.08.2022
11. Your clarification received vide email dated 01.09.2022
12. Tender Scrutiny Committee meeting held on 01.09.2022
13. TSC proceedings BESCOM/BC-19/GM(DAS)/2022-23/CYS-77 dated 01.09.2022
14. Letter No. BESCOM/BC-19/GM(DAS)/DGM-5/AGM-1/22-23/1212 dated 01.09.2022 informing about demonstration of SCADA-DMS-OMS functionalities
15. TSC proceedings based on the demonstration of SCADA-DMS-OMS functionalities held on 07.09.2022 issued vide BESCOM/BC-19/GM(DAS)/2022-23/CYS-83 dated 07.09.2022
16. Financial bids opened on 09.09.2022.
17. Resolution of 122nd Central Purchase Committee meeting held on 20.10.2022 issued vide BESCOM/BC-04/CS/AAO/A-1/F-1/2022-23/727E-727F dated 03.11.2022 – Subject No – 122/09
18. Your letter extending bid validity vide NCC/BESCOM/BC-19/2022-23/002 dated 17.12.2022
19. 1st Negotiation held in e-procurement portal on 06.01.2023
20. Your response vide NCC/BESCOM/BC-19/2022-23/003 dated 07.01.2023
21. Your letter No. NCC/BESCOM/BC-19/2022-23/005 dated 03.04.2023 extending EMD BG and bid validity
22. 2nd negotiation held in e-procurement portal on 14.06.2023
23. Your response vide letter No. NCC/Tenders/Electrical/2023-24/1041 dated 13.06.2023
24. Resolution of 123rd BESCOM BOD meeting held on 23.06.2023 subject No. 121/12
25. Your letter No. NCC/Tenders/Electrical/2023-24/1058 dated 25.07.2023 extending EMD BG validity
26. Letter of Intent No. BESCOM/BC-19/GM(DAS)/2023-24/713-714 dated 05.08.2023
27. Your letter of acceptance No. NCC/Tenders/Electrical/2023-24/1067 dated 09.08.2023
28. Price schedule breakup submitted vide email dated 28.08.2023
29. Bank Guarantee No. 1303923BG0000757 dated 30.08.2023 issued by State Bank of India valued at Rs.37,59,58,563 Valid up to 07.06.2030
30. BESCOM response to LOA vide letter No. BESCOM/BC-19/GM(DAS&SG)/2023-23/899 dated 05.09.2023
31. Contract Agreement signed on 16.09.2023

W. M. Shashidhar
11/10/23

The Detailed Work Award No. **BESCOM/GM(DAS)/2023-24/161** dated **11.10.2023** placed on your firm for "Upgradation of DAS in Bengaluru City" project with a contract period of six years and six months is herewith attached.

The Bid Document pertaining to subject tender enquiry forms a part of the Detailed Work Award. A copy of the bid document, amendments, addendums along with Schedule of Prices is enclosed.

This Detailed Work Award and enclosed documents are to be returned at the earliest duly signed over your firm's seal on all pages. The detailed work award number and date are to be quoted in all future correspondences to the office of the undersigned.


General Manager(DAS & SG)
BESCOM
W.M. SHASHIDHARA SHIVACHARYA 11/10/23
General Manager (DAS & TIC)

Copy submitted for kind information to:

1. Chief General Manager(I/A), Corporate Office, BESCOM, Bengaluru

Copy to:

1. PS to MD/D(T)/D(F) of BESCOM to place before Hon'ble MD, D(T)/D(F), BESCOM for information
2. AGM(F&C), DAS, BESCOM
3. MF/OC



BANGALORE ELECTRICITY SUPPLY COMPANY LIMITED
(Wholly owned Government of Karnataka Undertaking)

DETAILED WORK AWARD NO.: BESCOM/GM(DAS)/2023-24/161

DWA DATE: 11.10.2023

NAME OF THE WORK: UPGRADATION OF DAS IN BENGALURU CITY

I. SCOPE OF WORKS:

The scope of works in the project includes two main components mentioned below,

1. PART A - Upgradation of SCADA-DMS-OMS system.

Broadly, the scope of works pertaining to Upgradation of SCADA-DMS-OMS system design, engineering, development, configuration, FAT, supply, installation, SAT, training, documentation, commissioning, warranty and post warranty services of Distribution Automation System (DAS) Master Stations.

The scope also includes import /migration /integration of data from existing databases/systems /applications of DAS at both the control centers into the new system without loss of data and data integrity.

The scope includes integrating DAS Master Stations with the existing Communication System and providing design and engineering services to continue to perform SCADA operations in the event of modification/upgradation/expansion of the Communication System.

The existing interfaces with external systems like KPTCL's Master Control Center for SCADA, BESCOM's Centralized Customer Call Center System for call dockets and outage information exchange, ICT & MIS for GIS and CIS data, TIC section for BESCOM Mithra App, BESCOM Management Dashboard, Bengaluru Smart City Dashboard, etc. have to be integrated seamlessly with the new system. The system shall also provide APIs/Modules with necessary design and engineering services for integration with any other external systems in future at no cost to BESCOM.

The detailed scope of works pertaining to "Upgradation of SCADA-DMS-OMS System" is defined in the following documents in Section B of the Bid document cited in Ref [1] and its subsequent amendments and addendums cited in Ref [4] and [5].

1. **Section B-TS-01 - Introduction:** Provides general project background information. It also provides the list of clauses herein, which summarizes how the Technical Specifications for Upgradation of SCADA-DMS-OMS System housed in the control centers are organized.
2. **Section B-TS-02 - System Architecture:** Presents concepts and requirements with respect to the DAS hardware configuration, software and data architecture, configuration control, redundancy, failure management, availability, maintainability, and standards.
3. **Section B-TS-03 - SCADA:** Presents requirements that relate to the DAS supervisory control, data acquisition and data exchange functions.
4. **Section B-TS-04 - Information Storage and Information:** Presents the required DAS data archiving, retrieval, and reporting functions.
5. **Section B-TS-05 - User Interface:** Specifies the interface capabilities and features to be made available to users of the DAS. This includes the required login, security, jurisdiction, alarm, event, display, trending, web browser, and report features.
6. **Section B-TS-06 - Distribution Management System:** Specifies the distribution network modelling and application requirements of the DAS. The applications include Outage Management System, Switching Management System, Crew Management, Distribution State Estimation, Distribution Power Flow, Optimal Feeder Reconfiguration, Loss Minimisation, Feeder Fault Detection, Isolation and System Restoration, Detection of Open Conductor Faults, DERMS, Network Planning, Relay Co-ordination and Distribution Load Forecast. Relevant GIS interface requirements are also covered.
7. **Section B-TS-07 - Operator Training Simulator:** Specifies the capabilities and features to be provided with the Operator Training Simulator (OTS). This includes both the Trainer and Trainee requirements.
8. **Section B-TS-08 - Capacity and Performance:** Presents the specific capacity and performance requirements to be satisfied by the DAS. This includes the scenarios used to define the operational conditions under which these requirements shall be met.
9. **Section B-TS-09 - Hardware:** Identifies the DAS hardware and its requirements. It includes equipment such as servers, disks, backup and archiving devices, local and wide area network components, communication interfaces, firewalls, time and

frequency facilities, workstations, video projection systems, printers, enclosures, power supply units, and DAC simulators.

10. **Section B-TS-010 - Configuration & Development System, Quality Assurance System, and Support Software:** Presents requirements corresponding to the Configuration & Development System (C&DS) that will help familiarize Employer personnel with the DAS database, display, software, and report generation tools, and which later will be used to further develop and maintain the DAS. This includes the associated Quality Assurance System (QAS) that will be used to verify changes to the DAS prior to being placed in service. It also covers software requirements as relate to standardization, programming languages, database construction and maintenance, display generation and management, report generation, and software utilities.
11. **Section B-TS-011 - Documentation:** Specifies the documentation requirements, including the Review and Approval process, is to ensure that the documentation standard and coverage, when coupled with the delivered Training, is consistent with the aim of the Employer to be self sufficient in maintenance of the System.
12. **Section B-TS-012 - Quality Assurance and Testing:** Specifies various requirements associated with the Contractor's Quality Assurance Program, test plans, and test procedures. It also includes the variance recording, variance resolution, test scheduling, system inspection, factory acceptance test, and site acceptance test requirements.
13. **Section B-TS-013 - Training:** Details the Contractor's scope of work as it pertains to training Employer personnel. In this respect, it covers topics such as training documents, instructor qualifications, training curriculum, schedule, location, and number of students.
14. **Section B-TS-014 - Project Implementation:** Presents requirements as they relate to Contractor responsibilities, project organisation, project schedule, project documents, project security initiatives, system testing, shipment, and commissioning, system installation and startup, support services, system maintenance, spare parts, and test equipment.
15. **Appendix A - Glossary:** Provides a list of abbreviations and their meanings as used throughout these Technical Specifications.
16. **Appendix B - Capacity and Performance Data Tables:** Presents the specific data defined by the Employer as the basis for the Contractor to demonstrate that the DAS meets its capacity and performance requirements.

17. **Appendix C - Conceptual Diagrams:** Presents current block diagrams of the DAS and its associated Operations Department LAN, DAS Integration with External Systems and Server Rack Specifications.

Other tender documents pertaining to Upgradation of SCADA-DMS-OMS are,

1. Section A-GCC-General Conditions of Contract-SCADA-DMS-OMS
 2. Section A-SCC-Special Conditions Of Contract
 3. Section A-SFP-Standard Forms and Procedures-Upgradation of DAS
2. PART B - Supply, Installation, testing, commissioning and integration of Single panel Breaker (SPB) Type DAS RMU with Remote Terminal Unit (RTU) and required accessories at head end of the 11kV feeders in BBMP limits of Bangalore City

Broadly, the scope of works includes the following,

- Design, manufacture, supply, installation, commissioning and integration with DAS system of 11 kV class, SF6, SPB as per the Technical specification for 1808 No's of SPB with RTU
- Conducting tests on the 11 kV class, SF6, SPB and RTUs at manufacturer end as per IS & DAS specification.
- Packing, preservation, transportation from the manufacturer's work place to the consignee stores. Delivery including loading & unloading of materials at Consignee Stores.
- Transportation of SPB and RTUs to Employer specified site, Installation, Commissioning and conducting Integration test with DAS system.

The detailed scope of works pertaining to Single panel breaker is defined in the following documents in Section C of the Bid document cited in Ref [1] and its subsequent amendments and addendums cited in Ref [6] and [7].

1. **Section C-GTP-Guaranteed Particulars for RTU in SPB**
2. **Section C-GTP-Guaranteed Particulars for SPB**
3. **Section C-SPB-Implementation**
4. **Section C-TS-Technical Specification for RTU in SPB**
5. **Section C-TS-Technical Specification for SPB**

Other tender documents pertaining to Single panel breaker are,

1. Section A-GCC-General Conditions of Contract-SPB_RMUs-and-RTUs

2. Section A-SFP-Standard Forms and Procedures-Upgradation of DAS

Any other works not explicitly listed here but mentioned in the tender document cited in Ref [1] and required for implementing the functionalities described in the tender document shall be delivered by the firm, at no additional cost to BESCOM.

II. CONTRACTOR RESPONSIBILITIES

The contractor will be responsible for the following during the period of contract.

1. Part A – SCADA-DMS-OMS Upgradation

Contractor's specific responsibilities shall include:

- 1.1 Providing all System equipment and related support materials, including all interconnecting cables and wiring between all Contractor-provided equipment and between the System and any Employer designated interface points (Including but not limited to communication system and existing external interfaces).
- 1.2 Defining, supplying and maintaining the stock of spare parts needed to maintain system availability.
- 1.3 Providing all engineering, software design, development, and integration services necessary for System implementation.
- 1.4 Preparing all deliverable documentation for the delivery dates of each document (see Table 11-1 in Section B TS Clause 11 Documentation).
- 1.5 Providing and integrating all application software. This responsibility extends to all software developed by the Contractor's sub-contractors, if any.
- 1.6 Co-operating with all third parties (original equipment manufacturers or software sub-contractors used by the Contractor) in the resolution of problems that may be related to third-party software or equipment and which do not appear to function as specified. In particular, the Contractor shall be responsible for all third-party products that the Contractor uses as part of the System.
- 1.7 The Contractor shall be fully responsible for the preparation, verification and integration of the database into the new system. In consultation with the Employer, the Contractor shall develop a point naming convention that will be usable in an enterprise wide model of the Employers assets. The DAS system designed by the

Contractor must effectively handle the variations in point naming convention of the legacy system with respect to the new system during migration of the databases. The point naming convention shall account for other coincident projects such as a new GIS, Customer Call Centre System, Billing System, etc. to ensure that interoperability will be assured. The Contractor shall cooperate with third parties such as suppliers of these systems as needed to achieve the enterprise wide model.

- 1.8 The Contractor shall also be responsible for updating the database and displays prior to commissioning so that the system fully represents the actual electrical network at the time of commissioning. This includes installing the converted databases into the C&DS and Operator Training Simulator. In accord with this, the Contractor shall process and import the available databases that correspond to:
 - 1.8.1 Bengaluru City distribution network (GIS)
 - 1.8.2 RTU data as required to monitor and control the BMAZ
 - 1.8.3 RTU configurations
 - 1.8.4 KPTCL SCADA system data via TASE.2 data link as required to monitor and control substation devices within Bengaluru city
 - 1.8.5 Consumer Information and Asset Management System
 - 1.8.6 PGRS to receive information related to Customer Calls via MultiSpeak or equivalent protocol as decided by the Employer
 - 1.8.7 Weather data required for ADMS Application (historic and current weather data required for system operations shall be obtained by contractor throughout the contract period from reliable sources)
 - 1.8.8 Any other data that is needed to ensure a fully functioning DAS system.
- 1.9 The contractor shall be responsible to engage resources, provide hardware and software tools / licenses required to acquire the data from the existing databases or interface points indicated by the Employer.
- 1.10 The Contractor shall identify and enter any additional data elements required in the Contractor's database and shall in consultation with the Employer determine the correct values of those data elements. The mechanism for importation shall be repeatable so that the data within the NOM shall be able to be synchronised with changes that may be incorporated in the Employer's distribution network.

- 1.11 In Consultation with the Employer, the Contractor shall make an early delivery of C&DS and OTS for the Employer to familiarise, train its Operators and conduct integration testing of various interface systems.
- 1.12 The Contractor shall update patches and software updates during the entire Contract period including post-warranty period.
- 1.13 The Contractor shall perform periodic synchronisations. The Contractor shall undertake periodic synchronisation at least once before Pre-FAT and just before site commissioning activities. To support this check the Employer will support the Contractor by maintaining records of any changes to the existing network in accordance with reasonable guidelines developed in consultation with the Contractor.
- 1.14 In consultation with the Employer, developing a Display Style Guide outlying the principles of the display design and defining all discretionary display parameters used by the Contractor when developing displays and reports. The Contractor shall in the course of developing the style guide, develop a series of sample displays and reports for review and approval by the Employer.
- 1.15 In consultation with the Employer, defining operator roles, establishing logins, meeting security requirements, and designing the areas of jurisdiction that align with the Employer's operating procedures.
- 1.16 Developing display-naming conventions acceptable to Employer. In accordance with these conventions and the approved Display Style Guide, providing all DAS displays and reports, including those in Clause 5.14 of Section B-TS-05 - User Interface, as required to use and maintain the System. The Contractor shall create displays and reports to meet Employer approved layouts, standards, and formats. Prior to site commissioning activities, the Contractor shall recheck the displays to capture any changes that may have occurred from the initial creation. To support this check, the Employer will support the Contractor by maintaining records of any changes to the existing displays in accordance with reasonable guidelines developed in consultation with the Employer.
- 1.17 Customising standard System displays using the standards and conventions developed in consultation with the Employer.
- 1.18 Establishing the C&DS at Employer Site to allow the Employer to review and comment at its discretion on the Contractor's progress in the design and development of the System. The C&DS shall also be

used early by the Contractor to test and debug RTU protocol and interface issues, if any.

- 1.19 Training Employer personnel so that they will be self-sufficient and able to operate, maintain, and upgrade the complete System including the OTS.
- 1.20 Defining and coordinating a software and database management methodology that, from time to time, shall ensure synchronisation of the system databases and applications software between the Contractor and Employer sites until acceptance of the System. The C&DS shall be regularly updated with the latest versions of the database and displays for Employer review and comment. From the outset of the design phase, updates shall be scheduled at least every 2 months or whenever a significant development requires review by the Employer.
- 1.21 Periodically integrating the databases, displays, reports, and software defined and developed for the System by the Employer. As a minimum, such integration shall occur before the Factory Test and subsequently before the Site Test.
- 1.22 Ensuring that all software upon delivery is free of viruses, trapdoors, and other software contaminants, is purged of all sample scripts and sample code, and all default accounts and passwords removed or disabled. Ensuring complete compliance to Cyber Security rules and norms devised by the country and defined by the Employer by undergoing any additional necessary audits, testing and verification by standard bureaus as indicated by the Employer.
- 1.23 Managing, coordinating, and scheduling the activities of all sub-contractors employed by the Contractor for this project. This shall include the resolution of all problems that may arise in connection with the hardware, software, and services supplied by the sub-contractors.
- 1.24 Implementing the System according to quality standards acceptable to the Employer.
- 1.25 The Contractor is responsible and shall provide any additional licensing that may be required to interface and integrate updates with Employer's other IT/OT systems and other third party systems authorized by the Employer during the Contract period, irrespective of any upgradation that may happen in the Employer's IT/OT systems interfaced and/or integrated with the DAS system without any extra cost to BESCO.

- 1.26 Supplying documentation such as instruction manuals, maintenance manuals, drawings, software design and user documentation, and other appropriate material that together fully defines the System as delivered and commissioned and allows the Employer to operate, maintain, back up, restore, and upgrade the System hardware and software.
- 1.27 Keeping functional and design documents current.
- 1.28 Supplying final ("as built") documentation that is accurate and complete.
- 1.29 Performing the Preliminary Factory Test, resolve variances identified during Preliminary FAT. Report the Employer on readiness of the system to conduct Factory Acceptance Test.
- 1.30 Providing adequate facilities for Factory Acceptance Testing, providing adequate resources to conduct Factory Acceptance Testing, and performing the Factory Acceptance Testing (the Employer shall at its discretion be allowed to perform some or all of the actual test procedures under the supervision of the Contractor). Demonstrating a full-fledged capacity and performance of every functionality and interface of the DAS system as a whole as defined by the Employer.
- 1.31 Resolving all the variances reported during Factory Acceptance Test.
- 1.32 Transporting and delivering all Contractor-provided equipment and materials to the required Employer sites.
- 1.33 Providing the Employer with site implementation services (see Clause 14.6.3 of Section B TS Project Implementation).
- 1.34 Performing, with the Employer's assistance, system start-up after satisfactory system installation, i.e., powering up the system, loading correct versions of all software and databases, ensuring availability of communications networks, activating data links, verifying correct operation of the system, and verifying that the integrated system is ready for site testing.
- 1.35 Performing, with the Employer's assistance, after delivery and start-up of the system, but before any formal site testing, the setting up of all functions for proper operation (system and function "tuning").
- 1.36 Undertaking site acceptance testing at the Employer's site, including the correcting of all reported variances.

- 1.37 Undertaking point-to-point checkout with all field and external interfaces.
- 1.38 Provide tool for monitoring the functional and hardware availability of the system. Providing support during the Availability Testing phase. Resolving issues reported during Availability Test to meet the hardware and functional availability criteria defined by the Employer.
- 1.39 Ensuring and periodically demonstrating that the work is progressing according to the approved schedule.
- 1.40 Ensuring and periodically demonstrating at the discretion of the Employer that the hardware, software, and functionality, among other concerns, conform to the Contract.
- 1.41 Defining and providing the stock of spare parts, special equipment, and test equipment and tools needed to maintain the required System availability.
- 1.42 Providing adequate office space and services for Employer personnel at the Contractor's site.
- 1.43 Providing secure remote computer access from the Contractor's factory in order to support the installed systems.
- 1.44 Maintaining the System including the C&DS up to the start of the warranty.
- 1.45 Providing and implementing all required warranty services.
- 1.46 Providing all communication equipment, media, and interconnecting wiring from the existing hybrid (wired and wireless) communication system to the communication interfaces within the DAS to support SCADA requirements defined by the Employer.
- 1.47 Providing all communication equipment, media, and interconnecting wiring from the communications system to the communication interfaces within the DAS to support SCADA from proposed DNP-IP devices and integrate with existing CS Infrastructure.
- 1.48 Study the existing facilities including No Break Supply System, Video Project System, Air conditioning, Fire Suppression, Working Space (Furniture, Power outlets, Network Outlets, Server Racks) at both control centres and propose the system design such that these facilities are utilized optimally. Any additional facilities required over and above the existing facilities to integrate with the legacy system and to ensure longevity of the newly delivered system shall be supplied and commissioned by the contractor.

- 1.49 Provide rollout plan to phase out the legacy DAS system with new system with minimum interruption of commercial operations.
- 1.50 Provide a web-based tool for exchange of correspondence pertaining to project management, design, testing, supply, commissioning and overall execution of the project between the contractor and the employer. The tool shall be available from the date of issue of DWA and shall continue to remain after project completion.
- 1.51 Make arrangements for storeroom to house the hardware and tools prior to installation at Employer Site.
- 1.52 Throughout project implementation, cooperation and coordination with other contractors and Employer Personnel who will be working on the Bangalore DAS Upgradation Project.

To achieve an integrated system, the Contractor shall ensure among other things that:

- 1.53 The interface and protocol implementations cater for the differences that can arise intrinsically with different manufacturers of the new RTUs to be installed via other construction contracts comprising the Distribution Upgradation. In this respect, therefore, the resolution of any communication errors arising from protocol or interface discrepancies shall reside with the Contractor. In addition, the Contractor shall assume that the protocol may need specific parameterisation for the different models and brands of RTU.
- 1.54 The DAS must be tuned to complement the communication system such that transit times for data acquisition from the RTU including delivery to the screen at the Operator's console is the shortest possible.
- 1.55 The System cabling and its connections to all interface points such as power supply and communications equipment is complete and fit for purpose.
- 1.56 The existing interfaces and protocol implementations with external entities are tuned and configured to align with current requirements of the system.
- 1.57 The interfaces and protocol implementations with external entities that may be proposed in future are tuned and configured to align with any assumed requirements of the external entity.
- 1.58 All elements of the interoperability of the integrated system shall be error free.

- 1.59 All the data from the legacy system (including all the subsystems) must be migrated to the new system, ensuring data integrity and usefulness in the new system.
- 1.60 Unless otherwise specified in the Technical Specifications, the Contractor agrees to supply spare parts for a period of ten (10) years. The Contractor shall carry sufficient inventories to ensure an ex-stock supply of consumable spares for the plant and equipment. Other spare parts and components shall be supplied as promptly as possible, but at the most within three (3) months of placing the order and opening the letter of credit. In the event of termination of the production of spare parts, advance notification will be made to the Employer of the pending termination in sufficient time to allow the Employer to procure whatever spare parts may be deemed necessary. Following such termination, the Contractor shall furnish, to the extent possible and at no cost to the Employer, the blueprints, drawings and specifications of the spare parts, if requested.

2. Part B – SPB with RTU and Accessories

Contractor shall design, supply, install, commission and integrate the specified quantity of SPB with RTU in a given lot so that they are in good clean working order, fit for their intended purpose, and fully integrated with DAS master station. In this respect, the Contractor's responsibilities shall include, but shall not be limited to:

- 2.1 Ensuring the SPB with RTU are designed and engineered to meet the Employer's requirements fully and completely in coordination with the DAS master station
- 2.2 Supply all necessary materials and perform all necessary fabrication, testing, wiring, and interconnection work during the process of assembling or otherwise producing the SPB with RTUs.
- 2.3 Participation in project meetings, such as design and progress review meetings, and submission of all necessary transmittal related, for example, to items such as meeting agendas, meeting minutes, progress reports, etc.
- 2.4 Ensuring and periodically demonstrating that the work is progressing according to the approved schedule.
- 2.5 Submission of preliminary and as-built drawings, functional design documents, operation and maintenance manuals, test plans, test procedures, and test reports.
- 2.6 Timely Testing of Prototype SPB along with RTU at factory

- 2.7 Completing all routine testing on each SPB & RTU as well as factory acceptance testing (FAT) prior to delivering SPB and RTU batch to its assigned site. FAT shall include use of all necessary Contractor-supplied test equipment.
- 2.8 Hosting visits by Employer personnel to the manufacturing facilities so that these personnel can witness the SPB with RTU production and quality control procedures being implemented not at extra cost to Employer.
- 2.9 Provision of all necessary test equipment.
- 2.10 Transportation and installation of SPB with RTUs and all accessories to their assigned sites.
- 2.11 SAT shall include Contractor to demonstrate the integration of SPB and RTUs by end-to-end testing with the DAS. Operational Acceptance of site commences from Successful completion of SAT.
- 2.12 Training Employer staff so that they will be self-sufficient in monitoring, testing, and maintaining the SPB and RTUs equipment. This includes provision of all necessary training equipment and materials.
- 2.13 Maintaining the SPB and RTUs equipment until completion of the warranty period. Annual Maintenance contract will commence after completion of warranty period.
- 2.14 The Contractor has to abide by the prevailing standard safety policy.
- 2.15 Unless otherwise specified in the Technical Specifications, the Contractor agrees to supply spare parts for a period of ten (10) years. The Contractor shall carry sufficient inventories to ensure an ex-stock supply of consumable spares for the plant and equipment. Other spare parts and components shall be supplied as promptly as possible, but at the most within three (3) months of placing the order and opening the letter of credit. In the event of termination of the production of spare parts, advance notification will be made to the Employer of the pending termination in sufficient time to allow the Employer to procure whatever spare parts may be deemed necessary. Following such termination, the Contractor shall furnish, to the extent possible and at no cost to the Employer, the blueprints, drawings and specifications of the spare parts, if requested.

III. EMPLOYER RESPONSIBILITIES

Employer responsibilities in support of scope of work will include the following.

1. **Part A – SCADA-DMS-OMS**

- 1.1 Providing access to both control centres and associated facilities (as existing) including No Break Supply System, Video Project System, Air conditioning, Fire Suppression, Working Space (Furniture, Power outlets, Network Outlets, Server Racks).
- 1.2 Providing site access to Contractor personnel subject to undertaking any requisite training or certification for attendance on site. The Employer will provide or arrange for the required training or certification.
- 1.3 Providing documentation, interface information, engineering drawings, and schematic diagrams of the Employer furnished equipment to be directly interfaced with the Contractor supplied equipment, if needed.
- 1.4 Working with the Contractor to advise what information is necessary for the Contractor to implement the required functions of the System.
- 1.5 Indicate the existing databases and interface points to the contractor from which legacy data has to be imported by the contractor to the new system.
- 1.6 Providing data to the Contractor to support the Contractor's creation and population of all System databases as required for the System.
- 1.7 Working with the Contractor to advice and comment on display conventions and standards to be followed by the Contractor when preparing or customising displays for this project.
- 1.8 Reviewing and approving project deliverables such as, but not limited to, detailed implementation schedule, software and hardware functional design documents, user manuals, display and report standard formats, drawings, roll-out plan, progress reports, training program, quality assurance plan, test plans and procedures, test results, support services (including maintenance), and as-built system documents.
- 1.9 Providing TASE.2 and RTU information (including data and configuration details) for conversion to the System database and installation in the System.
- 1.10 Supporting the Contractor to develop the data not included in the available sources, but needed for the System database, where that data cannot be developed programmatically.

- 1.11 Providing documentation, interface details, engineering drawings, and schematic diagrams of the Employer-furnished equipment to be directly interfaced with the System.
- 1.12 Coordinating and supervising the Contractor's work to be performed at Employer facilities such as external equipment connection, site testing, etc.
- 1.13 Attending pre-factory tests (at the Employer's discretion).
- 1.14 Participating in factory tests, reporting variances and approving test results.
- 1.15 Reviewing and approving any Change Order Proposals that may be necessary.
- 1.16 Supervising the Contractor during System installation.
- 1.17 Participating in site tests, including point-to-point checkout, and approving test results.
- 1.18 Performing the availability test.
- 1.19 Providing test data for interfaces to systems external to the System such as other control centres.
- 1.20 Preparing variance reports, and retesting and approving corrected variances.
- 1.21 Determining if the Contractor's work is progressing in accordance with the schedule.
- 1.22 Verifying that all Contractor materials, installation practices, and workmanship conform to requirements.
- 1.23 Providing office/classroom space at the Employer's locations for the Contractor during system installation and pre-commissioning phase.

2. Part B – SPB with RTU and Accessories

Employer responsibilities will include supervision and support of all Contractor activities as follows:

- 2.1 Review and approval of Contractor-supplied drawings, manuals, design documents, training materials, test procedures, and other required documentation within 15 days from the date of submission.
- 2.2 Ensuring communications links between equipment and DAS master station.

- 2.3 Participation in Contractor-provided training courses.
- 2.4 Witnessing SPB with RTU production and quality control procedures as well as subsequent Prototype, factory and site acceptance tests.
- 2.5 Review and approval of Contractor-supplied work plans for installing and commissioning of SPBs and RTUs.
- 2.6 Inspection and approval of goods and materials both before and at time of delivery.
- 2.7 Coordination and supervision of the Contractor's on-site work.
- 2.8 Arranging Line Clear for installation and commissioning of SPB with RTUs.

IV. DATE OF COMMENCEMENT OF WORK:

The Contractor shall commence the work from the date of issue of this DWA. The date of issue of this DWA shall be treated as the Effective date of the contract.

V. PRICES:

The Contract Price of Rs.3,75,95,85,626/- (Rupees Three Hundred and Seventy Five Crores Ninety Five Lakhs Eighty Five Thousand Six Hundred and Twenty Six only) inclusive of all taxes and any other taxes applicable during the contractual period.

The prices are "FIXED PRICE" in pursuance to Section A GCC Clause 41 Contract Price. The following price schedules are annexed to the DWA as Annexure-1 which is the list of deliverables.

Price Schedule Details	Price Schedule Value inclusive of taxes
PRICE SCHEDULE I : PART-A SCADA-DMS-OMS HARDWARE AND SOFTWARE - SUPPLY PORTION	Rs. 92,77,15,666.06 inclusive of taxes (Rupees Ninety Two Crores Seventy Seven Lakhs Fifteen Thousand Six Hundred and Sixty Six and six paise only)
PRICE SCHEDULE I : PART B - SINGLE PANEL BREAKER TYPE DAS RMU WITH RTU AND OTHER	Rs. 2,33,94,96,552.98 inclusive of taxes (Rupees Two Hundred and Thirty Three Crores Ninety Four

ACCESSORIES – SUPPLY PORTION	Lakhs Ninety Six Thousand Five Hundred and Fifty Two and Ninety Eight paise only)
PRICE SCHEDULE II : PART-A SCADA-DMS-OMS IMPLEMENTATION – SERVICE PORTION	Rs. 11,61,52,384.32 inclusive of taxes (Rupees Eleven Crores Sixty One Lakhs Fifty Two Thousand Three Hundred and Eighty Four and Thirty Two paise only)
Price Schedule Details	Price Schedule Value inclusive of taxes
PRICE SCHEDULE II : PART B - SINGLE PANEL BREAKER TYPE DAS RMU WITH RTU AND OTHER ACCESSORIES – SERVICE PORTION	Rs. 17,05,26,200.55 inclusive of taxes (Rupees Seventeen Crores Five Lakhs Twenty Six Thousand Two Hundred and Fifty Five paise only)
PRICE SCHEDULE III : PART A- SCADA-DMS-OMS POST WARRANTY SERVICES	Rs. 12,27,92,794.20 inclusive of taxes (Rupees Twelve Crores Twenty Seven Lakhs Ninety Two Thousand Seven Hundred and Ninety Four and Twenty paise only).
PRICE SCHEDULE III : PART B- RMU and RTU POST WARRANTY SERVICES	Rs. 8,29,02,027.87 inclusive of taxes (Rupees Eight Crores Twenty Nine Lakhs Two Thousand and Twenty Seven and Eighty Seven paise only).

VI. CONTRACT PERIOD:

Unless otherwise terminated under the provisions of any other relevant clause,

➤ Contract period for PART-A Upgradation of SCADA-DMS-OMS includes the following:

- Implementation period of 18 months from the Effective date of Contract

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- ii. Defect Liability period of 24 months from the date of Operational Acceptance of the System
 - iii. Post-Warranty of 36 months after Defect Liability period.
- Contract period for PART-B: Supply, Installation, testing, commissioning and integration of Single panel Breaker (SPB) Type DAS RMU with Remote Terminal Unit (RTU) and required accessories at head end of the 11kV feeders in BBMP limits of Bangalore City
- i. Implementation period of 18 months from the Effective date of Contract
 - ii. Defect Liability period of 24 months from the date of Operational Acceptance
 - iii. Post-Warranty of 36 months after Defect Liability period.

VII. IMPLEMENTATION TIMELINES AND PAYMENT MILESTONES:

1. The terms of payment shall be based on payment milestones tabled below in accordance with contract values or actual costs that relate to Part A SCADA-DMS-OMS Upgradation and Part B Single Panel RMU with RTU and accessories.

PART A : SCADA-DMS-OMS Upgradation Milestones, Implementation Schedule and Payment

Mile stone No.	Milestone Name	Time Schedule to achieve the Milestone in Months starting from T. T being Effective date of contract	Value	Price Schedule Items Applicable
	Advance Payment	-	10%	Sch I PART A exclusive of taxes + Sch II PART A exclusive of taxes
1.MS1	Approval of final design documents and drawings	3	10%	Sch I, Part A exclusive of taxes+ Sch II Part A (Sl No 1+2+3) inclusive of taxes

Mile stone No.	Milestone Name	Time Schedule to achieve the Milestone in Months starting from T. T being Effective date of contract	Value	Price Schedule Items Applicable
1.MS 2	Successful completion of CDS and OTS FAT	4	5%	Sch I, Part A exclusive of taxes + Sch II Part A (Sl No 1+2+3) inclusive of taxes
1.MS 3	Successful Completion of FAT	7	20%	Sch I, Part A exclusive of taxes + Sch II Part A (Sl No 1+2+3) inclusive of taxes
1.MS 4	Hardware and Software Received at Control Centers	9		
	1.4.1 Hardware and Software Received at BCC1		10%	Sch I, Part A exclusive of taxes + Sch II Part A (Sl No 1+2+3) inclusive of taxes + 100% of taxes and duties on supplied items at BCC-1 + 50% of Price Schedule II Part A Sl No. 4 inclusive of taxes
	1.4.2 Hardware and Software Received at BCC2		10%	Sch I, Part A exclusive of taxes + Sch II Part A (Sl No 1+2+3) inclusive of taxes + 100% of taxes and duties on supplied items at BCC-2 + 50% of Price Schedule II Part A Sl No. 4 inclusive of taxes
1.MS 5	Issuance of Completion Certificate			
	1.5.1 Prorated Installation and Pre-commissioning of SCADA-DMS-OMS System at BCC1	10	7.5%	Sch I, Part A exclusive of taxes + Sch II Part A (Sl No 1+2+3) inclusive of taxes
	1.5.2 Prorated Installation and Pre-	11	7.5%	Sch I, Part A exclusive of taxes +

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	commissioning of SCADA-DMS-OMS System at BCC2			Sch II Part A (Sl No 1+2+3) inclusive of taxes
1.MS 6	Issuance of Operational Acceptance Certificate for complete SCADA-DMS-OMS system at both control centers	14	15%	Sch I, Part A exclusive of taxes + Sch II Part A (Sl No 1+2+3) inclusive of taxes
1.MS 7	Issuance of Final Acceptance Certificate for complete SCADA-DMS-OMS system at both control centers	18	15%	Sch I, Part A exclusive of taxes + Sch II Part A (Sl No 1+2+3) inclusive of taxes
2	Training – To be completed prior to achievement of MS 6. Payable on prorated basis based on the number of man-days of training provided	14		Sch II Part A Sl No 8 inclusive of taxes
3	Post Warranty Services – Payable half yearly subject to adherence of SLA described in Section B Clause 14 Implementation	3years after completion of Defect Liability Period		Sch III Part A Sl No 1+2 inclusive of taxes
4	Employer Project Expenses for Project Management Meeting is payable upon completion of each meeting as per actual number of Employer personnel attending the meeting			Sch II Part A Sl No 7 inclusive of taxes
5	Employer Project Expenses for C&DS and OTS FAT is payable upon successful completion of CDS and OTS FAT	4		Sch II Part A Sl No 5 inclusive of taxes

Mile stone No.	Milestone Name	Time Schedule to achieve the Milestone in Months starting from T. T being Effective date of contract	Value	Price Schedule Items Applicable
6	Employer Project Expenses for FAT is payable upon successful completion of FAT	7		Sch II Part A Sl No 6 inclusive of taxes
7	Design support training at Contractor's Facility payable after successful completion of Design support training	6		Sch II Part A Sl No 9 inclusive of taxes
8	Development and Deployment of Vehicle Management System (AVL)			Sch II Part A Sl No.10 inclusive of taxes

Note: Schedule I Part A is Supply Portion, Schedule II Part A is Service Portion and Schedule III Part A is Post Warranty Service.

PART B : Single Panel RMU with RTU and accessories- Milestones, Implementation Schedule and Payment

Milestone No.	Milestone Name	Timeline	Value	Price Schedule Items Applicable
	Advance Payment	-	10%	Sch I PART B exclusive of taxes + Sch II PART B exclusive of taxes
1.MS1 (SUPPLY)	Supply and Erection of RMUs and RTUs	Work to be completed in phases within 18 months from the	50%	50% of Price Schedule I Part B including 100% taxes and duties on supply and placing /installation of RMU and RTU including all accessories.

1.MS2 (SUPPLY)	Local testing of RMU with RTU	date of issue of work award	30%	30% of Price Schedule I Part B exclusive of taxes, upon connecting the RMU to the live distribution network and clearing Local Testing of the RMU with RTU.
1.MS3 (SUPPLY)	Integration with DAS Control Center		20%	20% of Price Schedule I Part B exclusive of taxes, upon clearing Operational acceptance (Integrating the remote site with DAS Control Center)
2.MS1 (SERVICE)	Local testing of RMU with RTU		60%	60% of Price Schedule II Part B inclusive of taxes, on connecting the RMU to the live distribution network and clearing local testing of RMU with RTU
2.MS2 (SERVICE)	Issuance of Operational Acceptance Certificate		40%	40% of Price Schedule II Part B inclusive of taxes, upon clearing Operational acceptance (Integrating the remote site with DAS Control Center)
3.MS3 (SERVICE)	AMC Services	3years after completion of Defect Liability Period		Sch III Part B SI No 1 inclusive of taxes. Payable on half-yearly basis after completion of preventive maintenance / repair work occurring during the half year period

Note: Schedule I Part B is Supply Portion, Schedule II Part B is Service Portion and Schedule III Part B is Post Warranty Service.

Invoices pertaining to Part B Milestones 1.MS3 Supply and 2.MS2 Service shall be raised for minimum of 250 Nos of Operational Acceptance Certificates.

- The advance payment mentioned in above tables will be made against receipt of invoice and an irrevocable advance payment security for the equivalent amount made out in favor of BESCOM as per Section A General Conditions of Contract Clause 43.2 Advance Payment Security and issued by a bank operating in India that is acceptable to

BESCOM. This advance shall be offset against the first three invoices submitted by the contractor. The validity of the Advance Payment Security submitted shall be extended as and when requested by BESCOM. The advance payment security shall be released after the offset of the advance payments made. The advance shall carry an interest prevailing in BESCOM.

3. Contractor shall submit the bill in approved proforma (one original and three copies) in accordance with the Contract Agreement and agreed milestones tabled above in this Clause VII to BESCOM enclosing necessary supporting documents listed below,
 - i. Copy of certificate of transit insurance
 - ii. Call letter for inspection from firm
 - iii. Dispatch Instructions issued by BESCOM
 - iv. Packing list or bill of materials
 - v. Contractor's detailed invoice
 - vi. Physical verification certificate by the site engineers.
4. Payment will be made as per conditions in this Detailed Work Award to the Contractor.
5. The prices mentioned should be all inclusive.
6. BESCOM shall verify the bill and supporting documents after receipt of bill from the Contractor. BESCOM shall release the payment after deducting applicable taxes, penalties/liquidated damages and other recoverable. All the payments made by BESCOM will be in Indian Rupee only. No payment made by BESCOM herein shall be deemed to constitute acceptance by BESCOM of the goods or services or any part(s) thereof.
7. If any excess payment has been made by BESCOM due to difference in quoted price in the bid and Contractor's invoice, BESCOM may without prejudice to its rights recover such amounts by other means after notifying the Contractor or deduct such excess payment from any payment subsequently falling due to the Contractor.
8. At its own discretion, BESCOM shall make payments either through crossed bank cheque or electronic bank transfer.
9. Wherever applicable, tax deduction certificates shall be issued at the end of the month against deduction of statutory taxes.
10. Payment will be arranged at the office of the undersigned by Assistant General Manager (F&C, DAS), BESCOM upon submission of invoices by the contractor duly certified by concerned officers and counter signed by General Manager(DAS&SG).

VIII. DEDUCTIONS FROM CONTRACT PRICE

All costs, damages or expenses which BESCOM may have paid, for which under the Contract the Contractor is liable, will be claimed by BESCOM. All such claims shall be billed by BESCOM to the Contractor regularly as and when they fall due. Such bills shall be supported by appropriate and certified vouchers or explanations, to enable the Contractor to properly identify such claims. Such claims shall be paid by the Contractor within thirty (30) days of the receipt of the corresponding bills and if not paid by the Contractor within the said period, BESCOM may then deduct the amount, from any money due or becoming due by BESCOM to the Contractor under the Contract or may be recovered by sections of Law or otherwise.

IX. TAXES, DUTIES AND LEVIES:

1. The GST and any other applicable Taxes shall be clearly spelt out in all transactions. If no mention is made then prices will be considered as inclusive of all taxes.
2. If there is any concession/reduction in the GST by the changes in the policies by the Central/State Government, the same shall be availed and passed on to BESCOM.
3. Any such Taxes, Duties, Levies etc., or rates not considered by the Contractor but payable as applicable on the date of billing shall be to Contractor's account and no separate claim on this behalf shall be entertained by BESCOM.
4. In case of deductions of statutory taxes, prevailing procedure will prevail.
5. As regards the Income Tax, Surcharge on Income Tax and any other Corporate Tax, BESCOM shall not bear any Tax liability whatsoever. The Contractor shall be liable and responsible for Payment of such Taxes as mandated under the provisions of the Law.
6. The Contractor shall ascertain from the concerned Tax Authorities of Government of Karnataka, the applicability of Work Contract Tax, Service Tax etc., in respect of this contract. No separate Claim in this regard will be entertained by BESCOM, as it is the responsibility of the Contractor to pay all these Taxes.
7. If there is any enhancement in the statutory taxes like GST, due to change in policies by the Central/State Government then such enhanced taxes will be payable by Contractor.
8. No claim for any increase towards statutory variation regarding enhancement of existing tax, GST or duty or introduction of a new tax or duty shall be entertained during the extended period of contract if the extension is required by causes attributable to the Contractor.

9. Except as otherwise specifically provided in the Contract, the Contractor shall bear and pay all taxes, duties, levies and charges assessed on the Contractor, or their employees by all municipal, state or national government authorities in connection with the Facilities in and outside of the country where the Site is located to complete the scope of works.
10. The Employer will reimburse duties and taxes up to no more than the actual amounts paid by the Contractor in the Employer's country. Such payments, however, shall be subject to the maximum duty and tax rates quoted in the Contractor's Price Bid. When requesting reimbursement, the Contractor shall furnish proof of the duties and taxes paid. Where applicable, the Employer will provide the Contractor with Form C.
11. If any tax exemptions, reductions, allowances or privileges may be available to the Contractor in the country where the Site is located, the Employer shall use its best endeavors to enable the Contractor to benefit from any such tax savings to the maximum allowable extent.
12. For the purpose of the Contract, it is agreed that the Contract Price specified in the Form of Contract Agreement is based on the taxes, duties, levies and charges prevailing at the date twenty-eight (28) days prior to the date of bid submission in India. If any rates of Tax are increased or decreased, a new Tax is introduced, an existing Tax is abolished, or any change in interpretation or application of any Tax occurs in the course of the performance of Contract, which was or will be assessed on the Contractor, or their employees in connection with performance of the Contract, an equitable adjustment of the Contract Price shall be made to fully take into account any such change by deduction therefrom, as the case may be, in accordance with Section A GCC Clause 26 Change in Laws and Regulations hereof.

X. PROGRESS REPORTS AND PHOTOGRAPHS

1. PROGRAM OF PERFORMANCE

Within twenty-eight (28) days after the date of signing the Contract Agreement, the Contractor shall prepare and submit to the Employer a detailed program of performance of the Contract, made in the form of GANTT and PERT charts used by commercially available software such as Microsoft Project or equivalent acceptable to the Employer and showing the sequence in which it proposes to design, manufacture, transport, assemble, install and pre-commission the Facilities, as well as the date by which the Contractor reasonably requires that the Employer shall have fulfilled its obligations under the Contract so as to enable the Contractor to execute the Contract in accordance with the program and

to achieve Completion, Commissioning and Acceptance of the Facilities in accordance with the Contract. The program so submitted by the Contractor shall accord with the Time Schedule defined in the Contract Agreement and any other dates and periods specified in the Contract. The Contractor shall update and revise the program as and when appropriate or when required by the Employer's Representative, but without modification in the Times for Completion given in the GCC Sub-clause 38 and any extension granted in accordance with Clause 33 Extension of Time for Completion of the contract agreement, and shall submit all such revisions to the Employer's Representative.

2. PROGRESS REPORT

The Contractor shall monitor progress of all the activities specified in the program referred to in Clause X Item 1 Program of Performance above, and supply a progress report to the Employer's Representative every month at its own cost with photographs etc. or as may be required by the Employer's Representative. The Contractor shall set up and maintain a project tracking system in the form of a project web site that will allow Authority, Contractor, and other authorized personnel (such as those associated with the project's other external interfaces/BESCOM's/KPTCL's other systems) to access and post information in the form of transmittals, reports, and other documents related to tracking and recording the status of the project's external interfaces and their various day-to-day activities.

The progress report shall be in a form acceptable to the Employer's Representative and shall indicate: (a) percentage completion achieved compared with the planned percentage completion for each activity; and (b) where any activity is behind the program, giving comments and likely consequences and stating the corrective action being taken.

3. PROGRESS OF PERFORMANCE

If at any time the Contractor's actual progress falls behind the program referred to in Clause X Item 1 Program of Performance, or it becomes apparent that it will so fall behind, the Contractor shall, at the request of the Employer or the Employer's Representative, prepare and submit to the Employer's Representative a revised program, taking into account the prevailing circumstances, and shall notify the Employer's Representative of the steps being taken to expedite progress so as to attain Completion of the Facilities within the Time for Completion under GCC Sub-Clause 38, any extension thereof entitled under Clause 33, or any extended period as may otherwise be agreed upon between the Employer and the Contractor.

XI. PROCEDURE AGAINST SUPPLY OF MATERIALS

1. The Contractor shall supply the materials/hardware/software (hereinafter referred as materials) required to complete the scope of works only after testing and Physical Inspection as per the technical specifications. The supply shall be made based on the delivery instruction issued by General Manager (DAS) duly adhering to prevailing governmental rules.
2. The Contractor shall be required to raise a call letter to General Manager, DAS as soon as materials are ready to arrange for inspection of materials so that the same are dispatched on approval.
3. The Contractor shall, at its own expense, handle all imported Plant and Equipment and Contractor's Equipment at the point(s) of import and shall handle any formalities for customs clearance.
4. The Contractor shall at its own risk and expense transport all the Plant and Equipment and the Contractor's Equipment to the Site by the mode of transport that the Contractor judges most suitable under all the circumstances.
5. The Contractor shall supply these materials and store the materials at his site. BESCOM Engineers shall verify and certify the supply, and provide the materials supplied for the works.
6. The Contractor shall arrange and supply the materials required.

XII. COMPLETION OF THE FACILITIES AND SITE ACCEPTANCE

The contractor shall carry out the works as defined in Clause 19 COMPLETION OF THE FACILITIES AND SITE ACCEPTANCE of the contract agreement to achieve completion of the facilities and site acceptance.

XIII. GUARANTEE/WARRANTY AND REPLACEMENT OF DEFECTIVE PARTS AND MATERIALS

1. PART-A: Upgradation of SCADA-DMS-OMS

- 1.1 The Contractor warrants that the Facilities or any part thereof shall be free from defects in the design, engineering, materials and workmanship of the Plant and Equipment supplied and of the work executed.
- 1.2 The Defect Liability Period, i.e., the warranty period, shall be twenty-four (24) months from the date of Operational Acceptance of the Facilities.

If during the Defect Liability Period any defect should be found in the design, engineering, materials and workmanship of the Plant and

Equipment supplied or of the work executed by the Contractor, the Contractor shall promptly, in consultation and agreement with the Employer regarding appropriate remedying of the defects, and at its cost, repair, replace or otherwise make good (as the Contractor shall, at its discretion, determine) such defect as well as any damage to the Facilities caused by such defect. The Contractor shall not be responsible for the repair, replacement or making good of any defect or of any damage to the Facilities arising out of or resulting from any of the following causes:

- (a) improper operation or maintenance of the Facilities by the Employer
- (b) operation of the Facilities outside specifications provided in the Contract
- (c) normal wear and tear.

1.3 The Contractor's obligations under this Clause XIII Item 1 shall not apply to

- (a) any materials that are supplied by the Employer under GCC Sub-Clause 15.2, are normally consumed in operation, or have a normal life shorter than the Defect Liability Period stated herein
- (b) any designs, specifications or other data designated, supplied or specified by or on behalf of the Employer or any matters for which the Contractor has disclaimed responsibility herein
- (c) any other materials supplied or any other work executed by or on behalf of the Employer, except for the work executed by the Employer as per Clause XIII 1.7 below.

1.4 The Employer shall give the Contractor a notice stating the nature of any such defect together with all available evidence thereof, promptly following the discovery thereof. The Employer shall afford all reasonable opportunity for the Contractor to inspect any such defect.

1.5 The Employer shall afford the Contractor all necessary access to the Facilities and the Site to enable the Contractor to perform its obligations under this Clause XIII Item 1. The Contractor may, with the consent of the Employer, remove from the Site any Plant and Equipment or any part of the Facilities that are defective if the nature of the defect, and/or any damage to the Facilities caused by the defect, is such that repairs cannot be expeditiously carried out at the Site.

1.6 If the repair, replacement or making good is of such a character that it may affect the efficiency of the Facilities or any part thereof, the Employer may give to the Contractor a notice requiring that tests of the defective part of the Facilities shall be made by the Contractor

immediately upon completion of such remedial work, whereupon the Contractor shall carry out such tests.

- 1.7 If such part fails the tests, the Contractor shall carry out further repair, replacement or making good (as the case may be) until that part of the Facilities passes such tests. The tests shall be agreed upon by the Employer and the Contractor.
- 1.8 If the Contractor fails to commence the work necessary to remedy such defect or any damage to the Facilities caused by such defect within a reasonable time (which shall in no event be considered to be less than fifteen (15) days), the Employer may, following notice to the Contractor, proceed to do such work, and the reasonable costs incurred by the Employer in connection therewith shall be paid to the Employer by the Contractor or may be deducted by the Employer from any monies due the Contractor or claimed under the Performance Security.
- 1.9 If the Facilities or any part thereof cannot be used by reason of such defect and/or making good of such defect, the Defect Liability Period of the Facilities or such part, as the case may be, shall be extended by a period equal to the period during which the Facilities or such part cannot be used by the Employer because of any of the aforesaid reasons.
- 1.10 Except as provided in Clause XIII Item 1 and GCC Sub-clause 23.1, the Contractor shall be under no liability whatsoever and howsoever arising, and whether under the Contract or at law, in respect of defects in the Facilities or any part thereof, the Plant and Equipment, design or engineering or work executed that appear after Completion of the Facilities or any part thereof, except where such defects are the result of the gross negligence, fraud, criminal or willful action of the Contractor.
- 1.11 If a Facility, or any part thereof, or the process performed thereby should require correction during the Defect Liability period, the Employer shall have the right to use such Facility, or part thereof, until such time as it is convenient to the Employer for the Facility, or part thereof, to be removed from service for correction. Such use shall not constitute acceptance nor shall it constitute a waiver of any of the Employer's rights under the Contract Agreement.
- 1.12 There shall be no restrictions on the Employer's rights to use, modify, expand, add to, or in any way adapt a Facility or any portion of it to best suit the Employer's needs as determined by the Employer at any time following the end of the Defect Liability Period. Any changes to be made by and in convenience of the Employer during the defects liability period, and the possible impact on warranty, will be mutually agreed in advance. The Employer has the further right to use the

Facility with or without additions for purposes not envisioned at the execution of the Contract Agreement and not described therein. These rights shall survive the Contract Agreement.

2. PART-B: Single Panel Breaker

- 2.1 The Contractor shall guarantee that all Equipment supplied by him will be new, unused and in accordance with the Contract Documents and free from defects in Material and Workmanship for a period of 24 Months from the date of issue of Operational acceptance by BESCOM. No repairs or replacement shall normally be carried out by BESCOM, when the Equipment is under the supervision of the Contractor's Supervisory Engineer.
- 2.2 Any defects noticed and/or failures/breakdown occurred during this period shall be rectified at free of cost to BESCOM. The repaired or new parts shall be supplied and erected at free of cost by the Contractor. Irrespective of number of failures and repairs, the contractor shall be responsible for free replacement of the defective materials supplied by the Contractor till the completion of Warranty period. If they are not rectified or replaced within this period, the contractor shall pay Liquidated Damages as per Liquidated Damages clause in the contract for the delay from the date of receipt of intimation for the defects or failures.
- 2.3 If any repair is carried out on contractor's behalf at the site, the Contractor shall bear the cost of such repairs. The cost of any special work required or general overhaul rendered necessary during the contract period due to defects in the material or defective work carried out by the Contractor shall be borne by the Contractor.
- 2.4 The incidental expenses, transport and freight charges for the replacement of defective materials within the contract period may also be borne by the contractor till such time it serves its stipulated contract period.
- 2.5 Site acceptance by BESCOM shall in no way relieve the Contractor of his obligations under guarantee clause.
- 2.6 The contractor shall guarantee among other things, the following
 - a) Quality and strength of materials used.
 - b) Safe electrical and mechanical stresses on all parts of the equipment under all specified conditions.
- 2.7 The provisions contained in this clause shall also be applicable in cases of normal wear and tear of the parts of the equipment supplied by the contractor.
- 2.8 If during the performance of the Contract, General Manager(DAS) shall decide and inform in writing/or through mail to the Contractor that the Contractor has furnished any equipment inferior to the quality specified, the Contractor on receiving notice of such defects or

deficiencies shall at its expense for making it good, furnish fresh equipment/materials up to the standards of the specifications as per BESCOM instructions. In case, the Contractor fails to do so, the Engineer may proceed to replace the equipment/materials so complained at the cost of the Contractor.

- 2.9 In the event of emergency where in the judgment of the Engineer, delay would cause serious loss or damages, repairs or adjustment may be made by the Engineer or a third party chosen by the Engineer without advance notice to the contractor and the cost of such work shall be paid by the contractor. In the event such action is taken by the Engineer, the Contractor will be notified promptly and he shall assist where possible in making necessary corrections. This shall not relieve the Contractor of his liabilities under the terms and conditions of the contract.
- 2.10 The Contractor's full and extreme liability under this clause shall be satisfied by the payment to BESCOM of extra cost, of such replacement procured including erection as provided for in the Contract, such extra cost being the ascertained difference between the price paid by BESCOM for such replacements and the Contract Price by portion for such defective equipment/materials/works and repayments of any sum paid by BESCOM to the Contractor in respect of such defective equipment/material. Should BESCOM not so replace the defective equipment/materials the Contractor's extreme liability under this clause shall be limited to repayment of all sums paid by BESCOM under the Contract for such defective equipment/materials.

XIV. CONTRACT PERFORMANCE GUARANTEE:

The contractor has submitted unconditional and irrevocable contract performance guarantee without any condition whatsoever in the form of a Bank Guarantee No. 1303923BG0000757 dated 30.08.2023 issued by State Bank of India at 10% of the contract price for an amount of Rs.37,59,58,563 (Rupees Thirty Seven Crores Fifty Nine Lakhs Fifty Eight Thousand Five Hundred and Sixty Three Only) Valid up to 07.06.2030 as performance security in the prescribed format. Performance Bank Guarantee shall remain valid for the entire contract period and three (3) months following the completion of the Contract Period i.e., the bank guarantee shall be valid for a period of 81 months from the date of issue of DWA.

The validity of BG submitted shall be extended up to completion of contract period as and when requested by BESCOM.

The performance bank guarantee shall be discharges/returned by BESCOM upon being satisfied that there has been no due in performance

of the obligations of the contractor under the contract. However no interest shall be paid by BESCO on the Bank Guarantee deposited.

In the event of the contractor being unable to service the contract for whatever reason, BESCO shall have the right to invoke the performance bank guarantee. BESCO shall notify the contractor in writing of the exercise of its right to receive such compensation within 14 days, indicating the contractual obligation(s) for which the contractor is in default.

BESCO is entitled to make recoveries from the contractor's bills, performance bank guarantee or any other payment due, in case of any error/omission on contractor's part.

XV. PENALTY AND LIQUIDATED DAMAGES

Penalty and liquidated damages shall apply as defined in Clause 15 PENALTY AND LIQUIDATED DAMAGES of the contract agreement reproduced below.

1.0 PART A – SCADA-DMS-OMS Upgradation

1.1 The Contractor guarantees that it shall attain Completion of the Facilities (or a part for which a separate time for completion is specified) within the Time for Completion specified in GCC Sub-Clause 38.2, or within such extended time to which the Contractor shall be entitled under Clause 33 Extension of time for completion of the contract agreement.

1.2 If the Contractor fails to attain the milestones by the time schedule specified in Clause VI and VII above, the Contractor shall pay to the Employer liquidated damages at one tenth of one percent (0.1%) of the milestone value per day of delay. If the Contractor completes the entire contract implementation within the time schedule specified in Clause VI and VII above, the liquidated damages applied for any previous milestones shall be withdrawn by the Employer.

1.3 No interest shall be paid to the Contractor for such amounts held as liquidated damages. The maximum liquidated damages are ten percent (10%) of the total Contract Price. Pursuant to Clause 34 Limitation of Liability of the contract agreement, the aggregate liability including liquidated damages shall in no event exceed twenty (20%) of the total Contract Price.

Once the "Maximum" is reached, the Employer may consider termination of the Contract, pursuant to Clause 31.1.2.2 of the

Contract Agreement. Such payment shall completely satisfy the Contractor's obligation to attain Completion of the Facilities or the relevant part thereof within the Time for Completion or any extension thereof under Clause 33 of the contract agreement. The Contractor shall have no further liability whatsoever to the Employer in respect thereof.

- 1.4 However, the payment of liquidated damages shall not in any way relieve the Contractor from any of its obligations to complete the Facilities or from any other obligations and liabilities of the Contractor under the Contract.
- 1.5 Save for liquidated damages payable under the Clause 30.1.2 of the Contract Agreement, the failure by the Contractor to attain any milestone or other act, matter or thing by any date specified in the Clause VII Implementation Timelines And Payment Milestones of this DWA and/or other program of work prepared pursuant to Clause X Item 1 Program of Performance shall not render the Contractor liable for any loss or damage thereby suffered by the Employer.
- 1.6 No bonus will be given if the Contractor attains Completion of the Facilities or any part thereof before the Time for Completion or any extension thereof under Clause 33 of the Contract Agreement.

2.0 PART B – SPB with RTU and Accessories

- 2.1 Penalty applicable during the contract period is described below.
 - 2.1.1 For delay in taking up the works as planned and not completing the installation of communication facility at remote sites, a penalty @ ½% per week will be levied against the works that are delayed, subject to maximum of TEN percentage (10%) of cost on the portion of work which is not completed
 - 2.1.2 The period of continuance of delay beyond 30 weeks shall be at the discretion of the Engineer. If the Contractor still fails to complete the work within such extended period, the Contract shall be cancelled.
 - 2.1.3 During the warranty period, if the contractor fails to provide replacement or service as defined, 10% of PBG cost on the portion of work which is not carried out will be deducted by BESCOM.
 - 2.1.4 In a nut shell, Penalties levied for the following:

- I. Non-fulfillment of program of works within the stipulated period.
 - II. The number of days delayed beyond the stipulated period for completing the work.
 - III. Extra cost that might be incurred by BESCO for getting the balance work completed through any other agency as it deems fit under the circumstances will be deducted from money due to him by BESCO for work done under the Contract or otherwise.
 - IV. Non Availability of Contractor Personnel during the execution of contract.
 - V. The total amount of Penalty for delay under the contract shall be subject to a maximum of ten percent (10%) of the contract price and BESCO has got every right to terminate the contract without any liability if it crosses 10%.
- 2.2 Liquidated damages applicable during contract period are as per Clause 17 Completion Time Guarantee of Section B GCC pertaining to SPB.

XVI. LIMITATION OF LIABILITY:

Except in cases of gross negligence or willful misconduct:

- a) The contractor shall not be liable to the Employer, whether in contract, tort, or otherwise, for any indirect or consequential loss or damage, loss of use, loss of production, or loss of profits or interest costs, provided that this exclusion shall not apply to any obligation of the Contractor to pay liquidated damages to the Employer and
- b) The aggregate liability of the contractor to the Employer whether under the contract, in tort or otherwise, shall not exceed twenty (20%) of total Contract Price, provided that this limitation shall not apply to the cost of repairing or replacing defective equipment, or to any obligation of the contractor to indemnify the Employer with respect to patent infringement.

XVII. TERMINATION OF CONTRACT

The conditions for termination of the contract are as described in Clause 31 TERMINATION OF CONTRACT of the contract agreement.

XVIII. SUSPENSION OF CONTRACT

The conditions for termination of the contract are as described in Clause 32 SUSPENSION of the contract agreement.

XIX. EXTENSION OF TIME FOR COMPLETION

1. The Contractor shall commence work on the Facilities from the Effective Date of Contract. The Completion of the Facilities shall be attained within the Implementation period as defined in Clause 6 CONTRACT PERIOD of this DWA from the Effective Date of Contract. The Time(s) for Completion specified shall be extended if the Contractor is delayed or impeded in the performance of any of its obligations under the Contract by reason of any of the following:
 - (a) any Change in the Facilities as provided in Clause 14.1 of the contract agreement
 - (b) any occurrence of Force Majeure as provided in Clause 27 of the contract agreement, unforeseen conditions as provided in Clause 29 of the contract agreement, or other occurrence of any of the matters specified or referred to in paragraphs (a), (b) and (c) of GCC Sub-Clause 22.2
 - (c) any suspension order given by the Employer under Clause 32 of the contract agreement or reduction in the rate of progress pursuant to Clause 32.2 of the contract agreement or
 - (d) any changes in laws and regulations as provided in GCC Clause 26 or
 - (e) any default or breach of the Contract by the Employer, specifically including failure to supply the items listed in the corresponding Appendix (Scope of Works and Supply by the Employer) to the Contract Agreement, or any activity, act or omission of any other contractors employed by the Employer or
 - (f) delays attributable to the Employer or caused by customs or
 - (g) any other matter specifically mentioned in the Contract

by such period as shall be fair and reasonable in all the circumstances and as shall fairly reflect the delay or impediment sustained by the Contractor.

2. Except where otherwise specifically provided in the Contract, the Contractor shall submit to the Employer's Representative a notice of a claim for an extension of the Time for Completion, together with particulars of the event or circumstance justifying such extension as soon as reasonably practicable after the commencement of such event or circumstance. As soon as reasonably practicable after receipt of

such notice and supporting particulars of the claim, the Employer and the Contractor shall agree upon the period of such extension. In the event that the Contractor does not accept the Employer's estimate of a fair and reasonable time extension, the Contractor shall be entitled to refer the matter to an Adjudicator, pursuant to Clause 40.1 of the contract agreement

3. The Contractor shall at all times use its reasonable efforts to minimize any delay in the performance of its obligations under the Contract.

XX. LIABILITY FOR ACCIDENTS AND DAMAGES

Under the Contract, the Contractor shall be responsible for loss or accidents which might arise owing to non – observation of safety requirements or else wise by its staff during erection of the Communication facility at remote or master sites. Also the contractor shall be responsible for loss or damage to the plant until successful completion of commissioning as defined elsewhere in the Bid document.

All demurrage, wharfage, and other expenses incurred due to delayed clearance of the Material or any other reason shall be to the account of the Contractor.

To deal with emergency/accidental eventualities at works site, the Contractor shall make all such arrangements necessary, such as services of an ambulance etc. for transportation to hospital at its cost.

XXI. BANKRUPTCY

If the contractor shall become bankrupt or otherwise insolvent or have a receiving order made against or compound with other creditors, or being a Corporation, commence to be wind up, not being a voluntary winding up for the purpose only of amalgamation/reconstruction, or carry on its business under a receiver for the benefit of its creditors or any of them, BESCO will be at liberty to terminate the contract forthwith by notice in writing to the contractor may become vested and BESCO reserves the right to encash the Contract Performance Guarantee.

XXII. DEFENCE OF SUITS

If any action in court is brought against BESCO or Engineer or an officer or agent of BESCO, for the failure, omission or neglect on the part of the Contractor to perform any acts, matters, covenants or things under the Contract, or for damage or injury caused by the alleged omission or negligence on the part of the Contractor, his agents, representatives or his Sub-Contractors, or in connection with any claim based on lawful demands

of Sub-Contractors, workmen, suppliers or employees, the Contractor shall in all such cases indemnify and keep BESCOM, and the Engineer and/or his representative, harmless from all losses, damages, expenses or decrees arising of such action.

XXIII. INSURANCE

All required insurances shall be taken by the contractor as per Clause 28 INSURANCE of the contract agreement.

XXIV. COMPLETENESS

All the necessary equipment, materials and software that are required for the work even if they have not been specifically mentioned in the contract specifications but are necessary for successful operation of the subject work are deemed to be included in the scope of the contractor.

XXV. FORCE MAJEURE

Force Majeure event or circumstance or combination of events and circumstances are as brought out in Clause 27 FORCE MAJEURE of the contract agreement.

XXVI. SETTLEMENT OF DISPUTES - LAW, JURISDICTION, DISPUTE RESOLUTION AND ARBITRATION

The details of applicable law, jurisdiction, dispute resolution and arbitration are as brought out in Clause 40 LAW, JURISDICTION, DISPUTE RESOLUTION AND ARBITRATION of the contract agreement.

XXVII. COMPLIANCE WITH LAWS AND REGULATIONS

The laws and regulations which are applicable to the Contractor and under which the Contract is to be constructed shall be Indian Laws and Regulations. The other terms and conditions regarding compliance with laws and regulations are as below:

The Contractor shall, in all matters arising in the performance of the Contract, comply with all respects, give all notices and pay all fees required by the provisions of any national or state statute, ordinance or other law or any regulation or by-law of any duly constituted authority.

The Contractor shall adhere to the statutory provisions under Payment of Minimum Wages Act, Contract Labor (Regulation & Abolition) Act, Employee's Provident Fund & Miscellaneous Provisions Act, Employees Compensation Act, Right to Information Act, and other relevant Statutes.

Non-compliance of the statutory provisions may attract penal action against Contractor from the Law Enforcing Authorities.

All liabilities arising out of the non-compliance of the Law of the Land by Contractor or its personnel will have to be borne by the Contractor, and the BESCOM will not be responsible in any manner whatsoever for the same.

The Contract shall in all respect be construed and interpreted in accordance with the Laws in force in India, including any such Laws passed or made or coming into force during the period of the Contract.

XXVIII. SEVERABILITY

Should any Provision(s) of this DWA be held invalid or unenforceable in any respect that provision shall not affect any other provisions and this DWA shall be construed as if the invalid or unenforceable provision had not been included.

XXIX. ENTIRE CONTRACT DOCUMENT

No modification, alteration or waiver of this DWA or any of its provisions shall be effective unless in writing and signed by both the parties.

XXX. The following documents shall constitute the Contract between BESCOM and the Contractor M/s NCC Limited, and each shall be read construed as an integral part of the Contract:

- a. Detailed Work Award (DWA)
- b. Contract Agreement
- c. Letter of Intent (LoI)
- d. Contractor's acceptance of Letter of Intent
- e. Bid and Price Scheduled submitted by Contractor
- f. Bid documents issued by BESCOM in GoK e-procurement portal vide Tender Ref No. BESCOM/BC-19/GM(DAS)/2022-23/07 dated 24.05.2022 (hereinafter called Tender) along with its amendments, addendums and corrigendum
 - i. Special Conditions of Contract
 - ii. General Conditions of Contract
 - iii. Bid data sheets
 - iv. Sample forms and procedures
 - v. Technical Specification, Drawings and Scope of works
 - vi. Pre-bid Queries and Clarifications
 - vii. Instructions to bidders
- g. Contract Performance Guarantee (CPG)
- h. Non-disclosure Agreement

Order of precedence:

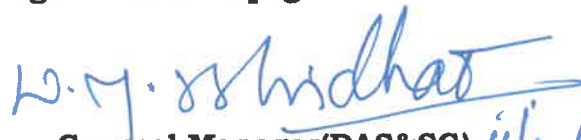
In the event of any ambiguity or conflict between the contract documents listed above, the order of precedence shall be the order in which the contract documents are listed in Clause XXX of the DWA.

XXXI. OTHER TERMS AND CONDITIONS

- The Contractor shall not assign or transfer the Contract or any right or interest therein in violation of the provision of this DWA. The contractor shall be responsible for guaranteed performance and timely completion of the total contract.
- All technical specification, and commercial terms and conditions as per the bid documents shall be complied with except for the deviations specifically agreed to.
- This Detailed Work award shall be read in conjunction with all the documents cited in the references of this DWA, along with subsequent Amendments, if any issued by this office.
- Any other incidental works/services which are not specifically mentioned in this DWA and are essentially required for the completion of the contract as per the bid documents along with subsequent amendments, if any issued by this office shall be executed by the Contractor at no additional cost to BESCOM.

All the terms and conditions cited in references of this DWA are applicable in addition to the terms and conditions stated in this DWA.

Please acknowledge the receipt of this Award within 7 days of receipt of this order and return this award duly signed on each page.


General Manager(DAS&SG) 11/10/20
BESCOM
W.M. SHASHIDHARA SHIVACHARYA
General Manager (DAS & TIC)

Copy submitted for kind information to:

- 1) Chief General Manager (Internal Audit) Corporate Office, BESCOM
- 2) Chief General Manager (Operations), Corporate Office, BESCOM

Copy to:

- 1) General Manager ICT & MIS, Corporate Office, BESCOM, Bengaluru
- 2) General Manager CR, Corporate Office, BESCOM, Bengaluru
- 3) SEE, PMU, Corporate Office, BESCOM, Bengaluru
- 4) Assistant General Manager, F&C, DAS, BESCOM
- 5) PS to D(T)/D(F)/MD, BESCOM to place before D(T)/D(F)/MD, BESCOM
- 6) MF/ OC.

PRICE SCHEDULE I : PART - A : SCADA-DMS-OMS HARDWARE AND SOFTWARE - SUPPLY PORTION

SCADA-DMS-OMS HARDWARE						
Sl No.	Item Description	Unit	Qty	Rates in INR	Amount in INR exclusive of taxes	Amount in INR inclusive of taxes
1	SCADA Server	No.	4	28,01,961.00	1,12,07,844.00	1,32,25,255.92
2	DMS Server	No.	4	22,82,140.00	91,28,560.00	1,07,71,700.80
3	OMS Server	No.	4	27,63,625.00	1,10,54,500.00	1,30,44,310.00
4	FEP server with interface switches	No.	4	14,52,271.00	58,09,084.00	68,54,719.12
5	ISR server	No.	4	53,94,151.00	2,15,76,604.00	2,54,60,392.72
6	NMS server	No.	4	6,22,402.00	24,89,608.00	29,37,737.44
7	DTS server	No.	1	28,33,775.00	28,33,775.00	33,43,854.50
8	ICCP Server	No.	4	14,09,806.00	56,39,224.00	66,54,284.32
9	Quality Assurance Server	No.	4	11,53,034.00	46,12,136.00	54,42,320.48
10	Operator Consoles (Three Monitor)	No.	16	5,22,818.00	83,65,088.00	98,70,803.84
11	Distribution Network Controller Console (Four Monitor)	No.	4	5,51,864.00	22,07,456.00	26,04,798.08
12	Operator Training Simulator Trainee (Three Monitor)	No.	2	5,22,818.00	10,45,636.00	12,33,850.48
13	Operations Department Interface Console (One Monitor)	No.	22	4,64,727.00	1,02,23,994.00	1,20,64,312.92
14	NMS Console (One Monitor)	No.	2	4,64,727.00	9,29,454.00	10,96,755.72
15	C&DS DAFE with Listening Mode Hardware	No.	2	14,22,728.00	28,45,456.00	33,57,638.08
16	QAS DAFE	No.	2	20,86,306.00	41,72,612.00	49,23,682.16
17	Developmental Server	No.	4	15,23,229.00	60,92,916.00	71,89,640.88

W.M. SINGH
General Manager / IIT
11/10

Sl No.	Item Description	Unit	Qty	Rates in INR	Amount in INR exclusive of taxes	Tax in INR @ 18%	Amount in INR inclusive of taxes
18	Web/Directory server	No.	4	12,43,658.00	49,74,632.00	8,95,433.76	58,70,065.76
19	Planning Consoles (Two Monitor)	No.	4	4,93,772.00	19,75,088.00	3,55,515.84	23,30,603.84
20	System Engineering Consoles (Two Monitor)	No.	6	4,93,772.00	29,62,632.00	5,33,273.76	34,95,905.76
21	Outage Co-ordinator Console (Two Monitor)	No.	2	4,93,772.00	9,87,544.00	1,77,757.92	11,65,301.92
22	C&DS Console (Two Monitor)	No.	8	4,93,772.00	39,50,176.00	7,11,031.68	46,61,207.68
23	Developmental console with one monitor	No.	4	4,64,727.00	18,58,908.00	3,34,603.44	21,93,511.44
24	DTS/Workstation Console with dual monitor	No.	1	4,64,727.00	4,64,727.00	83,650.86	5,48,377.86
25	External RAID Mass storage device (for 24 months online backup) - Storage Array with Historical Data Archival	No.	2	12,03,310.00	24,06,620.00	4,33,191.60	28,39,811.60
26	External DAT drive - Backup and Restoration of Servers	No.	2	62,24,020.00	1,24,48,040.00	22,40,647.20	1,46,88,687.20
27	Layer II switch (SCADA/DMS LAN)	Lot	2	39,41,880.00	78,83,760.00	14,19,076.80	93,02,836.80
28	Layer II switch (Planning & Development system LAN)	Lot	2	39,41,880.00	78,83,760.00	14,19,076.80	93,02,836.80
29	Router for interfacing IT system & SCADA/DMS DR centre	Lot	2	72,61,356.00	1,45,22,712.00	26,14,088.16	1,71,36,800.16
30	Web server with load balancing	No.	4	20,70,286.00	82,81,144.00	14,90,605.92	97,71,749.92
31	Mail server	No.	2	11,00,926.00	22,01,852.00	3,96,333.36	25,98,185.36
32	Firewall & network IDS/IPS	Lot	2	1,45,22,714.00	2,90,45,428.00	52,28,177.04	3,42,73,605.04
33	Layer II switch - DMZ	Lot	2	39,41,880.00	78,83,760.00	14,19,076.80	93,02,836.80
34	GPS Time synchronisation system	Set	4	4,14,935.00	16,59,740.00	2,98,753.20	19,58,493.20
35	Time, day & date digital displays	Set	2	4,14,935.00	8,29,870.00	1,49,376.60	9,79,246.60
36	Application Mirror Servers	Nos	4	27,63,625.00	1,10,54,500.00	19,89,810.00	1,30,44,310.00

General Manager (DAS & TDC)

Sl No.	Item Description	Unit	Qty	Rates in INR	Amount in INR exclusive of taxes	Tax in INR @ 18%	Amount in INR inclusive of taxes
37	Active Directory Server for DMZ integration with IT	Nos	2	11,00,926.00	22,01,852.00	3,96,333.36	25,98,185.36
38	File Servers	Nos	4	12,44,804.00	49,79,216.00	8,96,258.88	58,75,474.88
39	ISR Non System Server	Nos	2	53,94,151.00	1,07,88,302.00	19,41,894.36	1,27,30,196.36
40	Firewall for OFC Integration with existing CS	Nos	4	72,61,356.00	2,90,45,424.00	52,28,176.32	3,42,73,600.32
41	INMS Server for OFC Integration	Nos	2	2,17,841.00	4,35,682.00	78,422.76	5,14,104.76
42	INMS Console for OFC Integration	Nos	2	2,03,318.00	4,06,636.00	73,194.48	4,79,830.48
43	ISR DBI	Nos	2	25,98,688.00	51,97,376.00	9,35,527.68	61,32,903.68
44	GIS Proxy Server	Nos	1	19,62,401.00	19,62,401.00	3,53,232.18	23,15,633.18
45	Cabling	Lot	1	12,44,804.00	12,44,804.00	2,24,064.72	14,68,868.72
46	Remote PC Laptops	Nos	4	2,17,841.00	8,71,364.00	1,56,845.52	10,28,209.52
47	DAC Simulator PCs Laptops	Nos	8	2,03,318.00	16,26,544.00	2,92,777.92	19,19,321.92
	Mandatory spares						
48	Hard Drive for Storage Array of appropriate specification and capacity	Nos	16	1,24,480.00	19,91,680.00	3,58,502.40	23,50,182.40
49	Rackmount KB and Monitor	Set	2	1,86,721.00	3,73,442.00	67,219.56	4,40,661.56
50	KVM Switch and cable	Set	2	7,26,135.00	14,52,270.00	2,61,408.60	17,13,678.60
51	24" LED Monitor	Nos	4	27,468.00	1,09,872.00	19,776.96	1,29,648.96
52	Hard disk for Console / Workstation of appropriate specification and capacity	Nos	16	41,493.00	6,63,888.00	1,19,499.84	7,83,387.84
53	OFC	Mtr	500	622.00	3,11,000.00	55,980.00	3,66,980.00
54	OFC to Ethernet converter	Nos	10	1,03,734.00	10,37,340.00	1,86,721.20	12,24,061.20

Sl No.	Item Description	Unit	Qty	Rates in INR	Amount in INR exclusive of taxes	Tax in INR @ 18%	Amount in INR inclusive of taxes
55	OFC tool kit	Set	2	2,07,467.00	4,14,934.00	74,688.12	4,89,622.12
56	Cat 6 Ethernet Cable in meters	Mtr	500	622.00	3,11,000.00	55,980.00	3,66,980.00
57	Connectors for Ethernet Cable	Nos	50	1,037.00	51,850.00	9,333.00	61,183.00
58	Male-Female Converters for Ethernet Cable	Nos	20	6,225.00	1,24,500.00	22,410.00	1,46,910.00
59	Air filter for server racks and workstation	Nos	200	4,150.00	8,30,000.00	1,49,400.00	9,79,400.00
60	Door mount Rittal Server Rack Fan	Nos	4	12,416.00	49,664.00	8,939.52	58,603.52
61	Any other spare recommended by the Contractor**	L/s	1	-	-	-	-
	SCADA-DMS-OMS Hardware Sub-Total				29,99,89,881.00	5,39,98,178.58	35,39,88,059.58

SCADA-DMS-OMS SOFTWARE

Sl No.	Item Description	Unit	Qty	Rates in INR	Amount in INR exclusive of taxes	Tax in INR	Amount in INR inclusive of taxes
62	SCADA software	Lot	1	16,74,62,027.00	16,74,62,027.00	3,01,43,164.86	19,76,05,191.86
63	ISR Software	Lot	1	2,88,42,249.00	2,88,42,249.00	51,91,604.82	3,40,33,853.82
64	DMS software	Lot	1	2,39,08,924.00	2,39,08,924.00	43,03,606.32	2,82,12,530.32
65	OMS Software	Lot	1	1,95,61,848.00	1,95,61,848.00	35,21,132.64	2,30,82,980.64
66	DTS software	Lot	1	1,30,41,231.00	1,30,41,231.00	23,47,421.58	1,53,88,652.58
67	Network Management Software	Lot	1	89,69,306.00	89,69,306.00	16,14,475.08	1,05,83,781.08
68	RDBMS package	Lot	1	2,08,48,797.00	2,08,48,797.00	37,52,783.46	2,46,01,580.46


W.M. SHASHINDHARA SHIVACHARYA
 General Manager (DAS & TIC)

Sl No.	Item Description	Unit	Qty	Rates in INR	Amount in INR exclusive of taxes	Tax in INR @ 18%	Amount in INR inclusive of taxes
69	GIS Adaptor /Engine for importing data from GIS system under IT system	Lot	1	1,00,74,046.00	1,00,74,046.00	18,13,328.28	1,18,87,374.28
70	Other 3rd party software with necessary licenses like ESRI, Autodesk, Storage array management applications, Hypervisors, Oracle, OS , Antivirus, NPM, Domain controller, DAQ Simulator Software, etc	Lot	1	6,88,13,408.00	6,88,13,408.00	1,23,86,413.44	8,11,99,821.44
71	INMS Monitoring License	Nos	6000	20,698.00	12,41,88,000.00	2,23,53,840.00	14,65,41,840.00
72	Automatic Vehicle Location Server along with OS and Software Licenses (Refer to Pre Bid Clarification point No.27==> Procurement of AVL is in scope of BESCOM, Integration of AVL system with proposed application is in Scope of Bidder. Hence we are quoting price for only Integration)	Nos	2	2,50,000.00	5,00,000.00	90,000.00	5,90,000.00
73	Any other software to meet functional /performance requirement	Lot	1	-	-	-	-
SCADA-DMS-OMS Software Sub-Total					48,62,09,836.00	8,75,17,770.48	57,37,27,606.48
SCADA-DMS-OMS Hardware and Software Total Inclusive of taxes					78,61,99,717.00	14,15,15,949.06	92,77,15,666.06


W.M. SHASHIDHARA SHIVACHARYA
 General Manager (DAS & TIC)

PRICE SCHEDULE II : PART - A : SCADA-DMS-OMS IMPLEMENTATION - SERVICE PORTION

SI No.	Item Description	Unit	Qty	Rates in INR	Amount in INR exclusive of taxes	Tax in INR @18%	Amount in INR inclusive of taxes
1	Design & Engineering Services	Lot	1	5,42,91,258.00	5,42,91,258.00	97,72,426.44	6,40,63,684.44
2	Installation of equipment and systems at all sites, equipment and system testing, etc.,	Lot	1	1,66,38,132.00	1,66,38,132.00	29,94,863.76	1,96,32,995.76
3	Integration with legacy applications & Data Migration	No.	1	96,32,603.00	96,32,603.00	17,33,868.54	1,13,66,471.54
4	Local Transportation, Insurance & Erection for one Control Center	Lot	2	5,10,616.00	10,21,232.00	1,83,821.76	12,05,053.76
SUB - TOTAL FOR SCADA-DMS-OMS IMPLEMENTATION SERVICE PORTION							
5	Employer Project Expenses for attending C&DS and OTS FAT for five days for four persons - at Contractor's Facilities	No.	1	17,99,417.00	17,99,417.00	3,23,895.06	21,23,312.06
6	Employer Project Expenses for attending FAT for 6 Weeks for four persons - at Contractor's Facilities	No.	1	24,19,115.00	24,19,115.00	4,35,440.70	28,54,555.70
7	Employer Project Expenses for attending Project Management meetings - 4 personnel attending meetings for 5 days at Contractor's Facilities	No.	5	3,66,587.00	18,32,935.00	3,29,928.30	21,62,863.30
8	Training for SCADA-DMS-OMS System including Course materials, equipment, presentation, etc.,	Lot	1	37,93,999.00	37,93,999.00	6,82,919.82	44,76,918.82
9	Design Support Training at Vendor's facility	No.	1	67,55,533.00	67,55,533.00	12,15,995.94	79,71,528.94
10	Development and Deployment of Vehicle Management System (AVL)(Refer to Pre Bid Clarification point No.27==> Procurement of AVL is in scope of BESCOM, Integration of AVL system with proposed application is in Scope of Bidder. Hence we are quoting price for only Integration)	No.	1	2,50,000.00	2,50,000.00	45,000.00	2,95,000.00
TOTAL FOR SCADA-DMS-OMS IMPLEMENTATION SERVICE PORTION					9,84,34,224.00	1,77,18,160.32	11,61,52,384.32

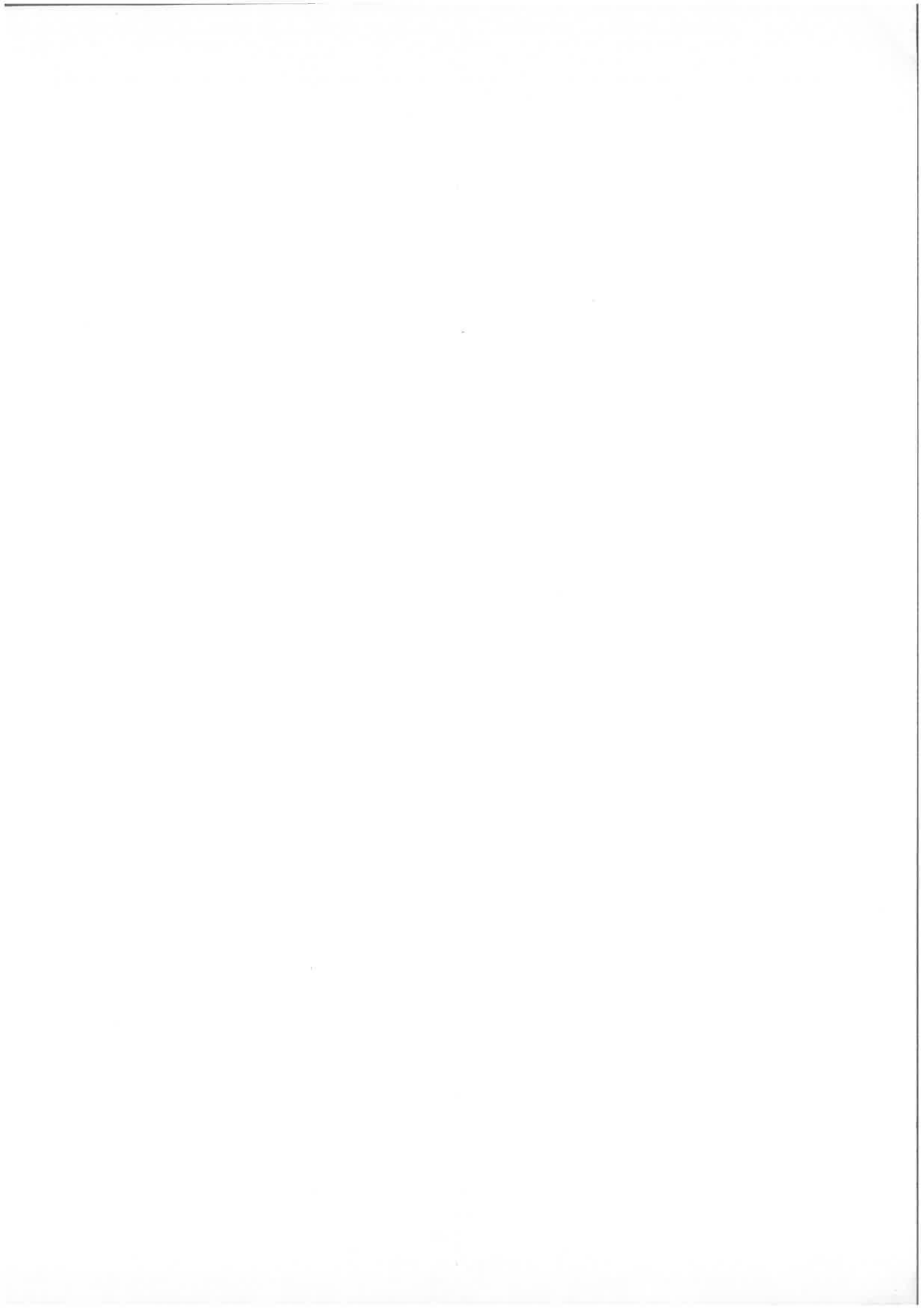

W.M. SHASHIDHARA SHRINATHARYA
 General Manager (DAS & TIC)

PRICE SCHEDULE III : PART - A : SCADA-DMS-OMS IMPLEMENTATION POST WARRANTY SERVICES

Sl No.	Item Description	Unit	Qty	Rates in INR	Amount in INR exclusive of taxes	Tax in INR 18%	Amount in INR inclusive of taxes
1	Post-Warranty Services for one year - Hardware Maintenance	No.	3	87,17,845.00	2,61,53,535.00	47,07,636.30	3,08,61,171.30
2	Post-Warranty Services for one year - Software Maintenance	No.	3	2,59,69,385.00	7,79,08,155.00	1,40,23,467.90	9,19,31,622.90
	TOTAL FOR SCADA-DMS-OMS POST WARRANTY SERVICES				10,40,61,690.00	1,87,31,104.20	12,27,92,794.20


W.M. SHASHIDHARA SHIVACHARYA
 General Manager (DAS & TIC)

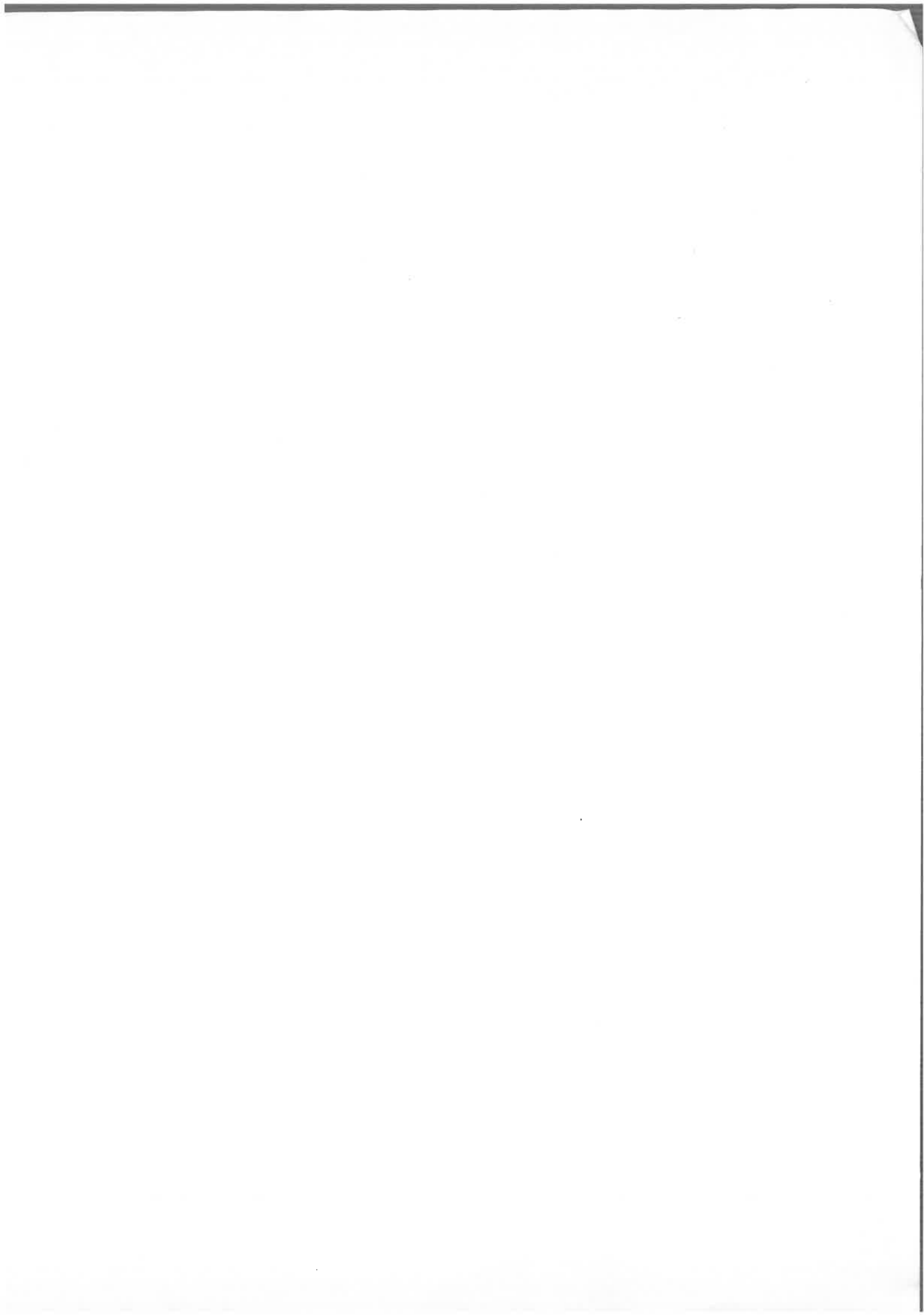
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PRICE SCHEDULE I : PART - B : SINGLE PANEL BREAKER TYPE DAS RMU WITH RTU AND OTHER ACCESSORIES - SUPPLY PORTION

Sl No.	Item Description	Unit	Qty	Rates in INR	Amount in INR exclusive of taxes	Tax in INR	Amount in INR inclusive of taxes
1	Single Panel Breaker Type RMU with Incoming and Outgoing including accessories for DAS and RTU hosting	No	1,808.00	7,72,021.00	1,39,58,13,968.00	25,12,46,514.24	1,64,70,60,482.24
2	Bed Concreting with CC 1:2:4 as per Labour Charges	Cmt	253.12	9681	24,50,454.72	4,41,081.85	28,91,536.57
3	Foundation with Reinforcement CC 1:2:4 as per Labour Charges	Cmt	1,048.64	15589	1,63,47,248.96	29,42,504.81	1,92,89,753.77
4	Providing Reinforcement as per Labour Charges	Kg	36,160.00	131	47,36,960.00	8,52,652.80	55,89,612.80
5	Supplying of Frame of MS Channel 100x50mm and MS angle 40x40x5mm welding etc., complete.	Kg	43,392.00	150.00	65,08,800.00	11,71,584.00	76,80,384.00
6	Rod type earthing with 40 mm Dia, 3 Mtr long MS rod as per specification & drawing (enclosed)	No	3,616.00	1,799.00	65,05,184.00	11,70,933.12	76,76,117.12
7	Lettering the RMU with enamel paint and also writing single line diagram of each panel, caution Board, Danger Board etc., including cost of Paint, Brush etc.,	No	1,808.00	1,535.00	27,75,280.00	4,99,550.40	32,74,830.40
8	11 kV, 3x400 Sqmm XLPE UG Cable Using Conventional Laying with Excavation (Laying of 20mtr length)	Km	36.16	28,85,711.00	10,43,47,309.76	1,87,82,515.76	12,31,29,825.52
9	Cable Termination Kit Indoor HS Type Suitable for 400 Sqmm Cable	Per Kit	3,616.00	3,748.00	1,35,52,768.00	24,39,498.24	1,59,92,266.24
10	Straight Through Jointing Kits HS Type suitable for 400 Sqmm Cable	No	3,616.00	10,119.00	3,65,90,304.00	65,86,254.72	4,31,76,558.72
11	RTU with firmware and functionalities/ capabilities as described in the technical specification including installation, testing and commissioning	No	1,808.00	2,17,365.00	39,29,95,920.00	7,07,39,265.60	46,37,35,185.60
Single panel breaker type DAS RMU with RTU and other accessories - SUPPLY TOTAL Inclusive of taxes					1,98,26,24,197.44	35,68,72,355.54	2,33,94,96,552.98

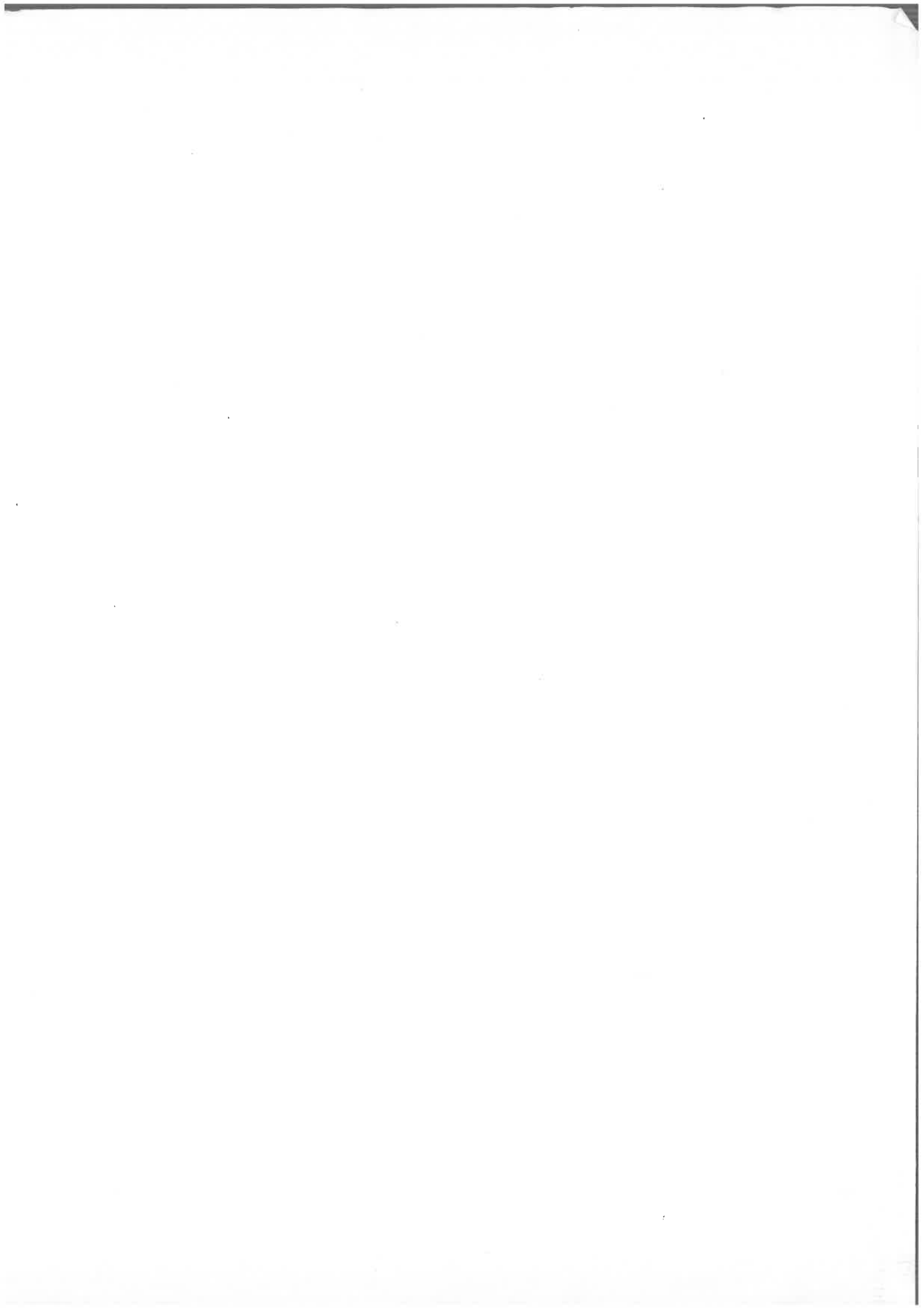

W.M. SHASHIDHARA SHIVACHARYA
 General Manager (DAS & TIC)



PRICE SCHEDULE II : PART - B : SINGLE PANEL BREAKER TYPE DAS RMU WITH RTU AND OTHER ACCESSORIES - SERVICE PORTION

Sl No.	Item Description	Unit	Qty	Rates in INR	Amount in INR exclusive of taxes	Tax in INR 18%	Amount in INR inclusive of taxes
1	Earth Excavation for RMU Foundation depositing of earth on bank up to a lead of 50 Mtr and with a lift up to 1.5 Mtr- Ordinary Soil	Cmt	2,061.12	300.00	6,18,336.00	1,11,300.48	7,29,636.48
2	Fixing foundation frame of MS channel 100x50 mm and MS angle 40x40x5 mm welding, fixing in concrete, aligning the RMU on foundation bed, assembly of units, connecting Bus Bars from panel to panel etc., complete.	No	3,616.00	12,367.00	4,47,19,072.00	80,49,432.96	5,27,68,504.96
3	11 kV, 3x400 Sqmm XLPE UG Cable Using Conventional Laying with Excavation (For laying of 20mtr Length)	Km	36.16	11,16,208.00	4,03,62,081.28	72,65,174.63	4,76,27,255.91
4	Making and Fixing pot head for HT Cable 3x400 Sqmm	No	3,616.00	2,399.00	86,74,784.00	15,61,461.12	1,02,36,245.12
5	Heat shrinkable Straight Through Joint 3x400 Sqmm	No	3,616.00	5,996.00	2,16,81,536.00	39,02,676.48	2,55,84,212.48
6	Installation, testing and commissioning of RTU	No	1,808.00	15,740.00	2,84,57,920.00	51,22,425.60	3,35,80,345.60
TOTAL FOR SINGLE PANEL BREAKER TYPE DAS RMU WITH RTU AND OTHER ACCESSORIES - SERVICE PORTION					14,45,13,729.28	2,60,12,471.27	17,05,26,200.55

W.M. SHASHIDHARA SHIVACHARYA



PRICE SCHEDULE III : PART - B : SINGLE PANEL BREAKER TYPE DAS RMU WITH RTU AND OTHER ACCESSORIES POST WARRANTY SERVICES

Sl No.	Item Description	Unit	Qty	Rates in INR	Amount in INR exclusive of taxes	Tax in INR @18%	Amount in INR inclusive of taxes
1	Post-Warranty Services for one year for Single Panel Breaker Type DAS RMU with RTU and other Accessories	No.	3	2,34,18,651.94	7,02,55,955.82	1,26,46,072.05	8,29,02,027.87
	TOTAL FOR SINGLE PANEL BREAKER TYPE DAS RMU WITH RTU AND OTHER ACCESSORIES - POST WARRANTY SERVICES				7,02,55,955.82	1,26,46,072.05	8,29,02,027.87


W.M. SHASHIDHARA SHIVACHARY,
 General Manager (DAS & TIC)

11/10/25

